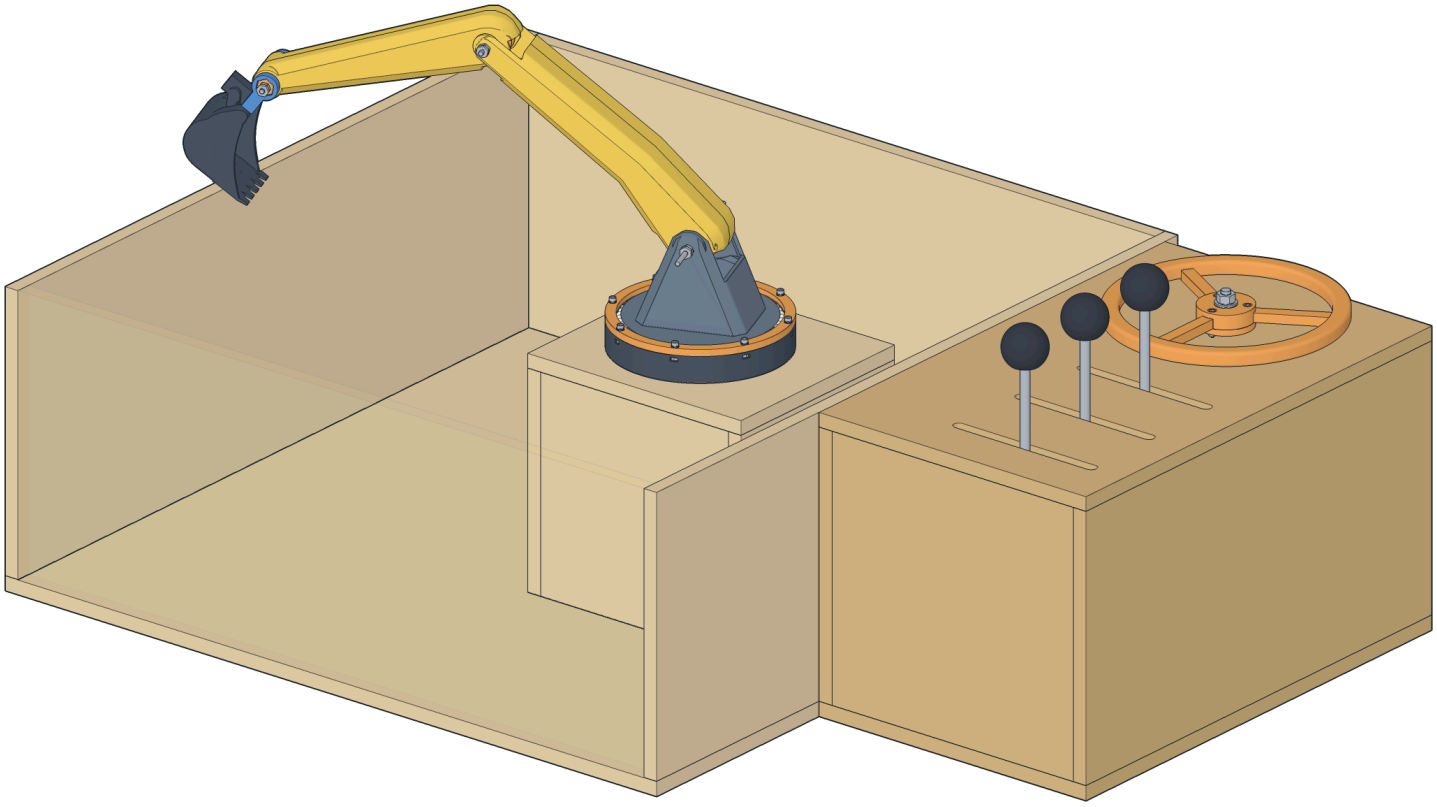
ASSEMBLY & CONFIGURATION MANUAL

EX-MA CABLE EXCAVATOR

Simplified EX-MA arm · lever control box · pedestal · sandbox



Edition 2026-10-10 · for the project files in `2026-09-24-EX-MA-simplified/`
Includes the 2026-09-29 base and bucket-ear rebuilds · all illustrations rendered from the project CAD

26 printed part types · 29 prints
21 plywood panels · 4 rope circuits · no electronics

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Page numbers count the cover as page 1. Figures are numbered in order through the manual.

BEFORE YOU BEGIN

ABOUT THIS MANUAL

This manual takes you from a table of printed parts, plywood and hardware to a working, rigged excavator. Work through the stages in order: each one assumes the stages before it are complete and checked.

HOW THE PAGES WORK

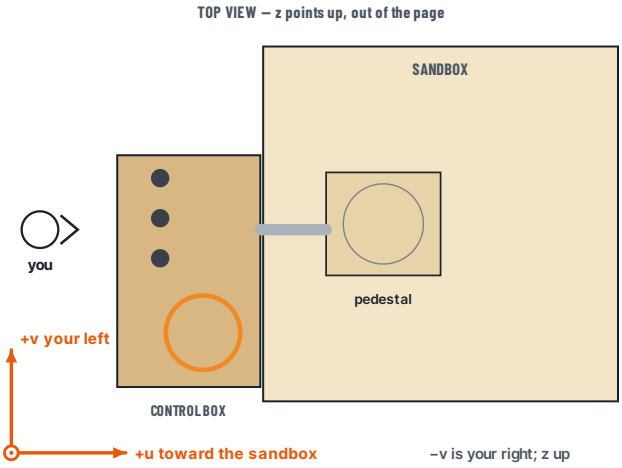
- **Stage header** — the stage number and what it achieves.
- **You need** — every part and fastener for that stage, with its ID (P05 printed, W01 wood, H03 hardware, S04 stock) and quantity. Lay them out first.
- **Numbered steps** — do them in order; each step is one action.
- **Figures** — numbered balloons (1) match the step text or the list beside the figure. **Orange arrows** show where a part moves to.
- **Check** boxes — stop and verify before going on.

DIRECTIONS USED EVERYWHERE

The project uses one fixed frame. Picture yourself standing at the control box, facing the sandbox:

- **+u** points away from you, toward the sandbox (“forward”).
- **+v** points to your left; **–v** to your right.
- **z** points up.

Part names use the same words: the *left* boom half is the +v half; the sandbox wall “–u” is the wall the box bolts to.



SYMBOLS

WARNING Risk of injury. Read before doing the step.

CAUTION Risk of damaging a part or having to take things apart again.

COMMON MISTAKE Something that is easy to get wrong at this point.

CHECK A verification point. Do not continue until it passes.

TIP Practical advice. Helpful, but not taken from the CAD model.

OPEN ITEM A question this manual found in the design files that you must decide on.

VERIFIED, DERIVED OR ASSUMED?

Every number and connection in this manual comes from the project files. Where a statement needs care, it carries a tag:

Tag	Meaning
VERIFIED	Read directly from the CAD model, the build scripts or their validation reports.
DERIVED	Worked out from the model geometry for this manual (for example an assembly order that the geometry forces). The reasoning is given next to it.
ASSUMPTION	Not in the project files: general practice or a choice you can change. Treat it as a suggestion.
OPEN ITEM	A conflict or gap found in the design files. Appendix D lists all of them with options.

Colours in the illustrations code the part type, not the filament colour: printed PLA+ part · arm shell (PETG) · bucket ears · original EX-MA dark parts · plywood · metal hardware. Ropes: boom · stick · bucket · slew. Parts already installed in earlier steps are drawn in light grey.

READ FIRST

SAFETY AND GOOD PRACTICE

The EX-MA is a hand-powered model: there are no motors, springs or electrics. The hazards are the tools you build it with, cut wire rope, small parts and pinch points between moving printed parts.

CUT WIRE ROPE IS SHARP. Each cut end of 7×19 rope splays into fine, stiff wires that puncture skin. Wrap tape round the rope before cutting and cut through the tape; wear gloves when handling free ends; wear eye protection when cutting.

SMALL PARTS. The 6 mm BBs, M3 nuts and crimp sleeves are choking hazards. The M3 kit is labelled “not suitable for children under 3 years due to small parts”. Keep the build area and the finished model away from small children.

PINCH POINTS. The slewing ring, the lever hubs and every joint drum can trap fingers while ropes are tensioned. Keep fingers out of the box and the pedestal while someone moves a lever or the wheel.

POWER TOOLS AND BLADES. Clamp plywood before drilling; cut PTFE tube with a tube cutter or a new razor blade on a cutting board, never toward your hand.

ADHESIVES. CA glue bonds skin in seconds. Use it with ventilation and keep acetone-free remover nearby. Follow the maker’s instructions and cure times for wood glue, CA, epoxy and silicone.

HEAT. Printed PLA+ softens well below the temperature of a car interior in summer. Do not leave the model in direct sun or in a hot vehicle. ASSUMPTION

DO NOT FORCE A JOINT. Every lever has a hard stop at the end of its slot in the box top. If a lever or the wheel resists before reaching a stop, find the cause (page “Troubleshooting”) instead of pushing harder; a rope can pull out of its crimp or a printed drum can crack.

SAND. Sand in the slewing ring, the lever slots or the conduit wears the parts fast. Keep the sand dry and below the pedestal top (the model’s sand level is 10 mm below it VERIFIED).

WORKSPACE

- A flat table about 1.2 × 0.8 m for the box and the sandbox; good light (rope holes are Ø2.4 mm).
- Small trays or bags for hardware: one per stage (the “You need” lists match the stages).
- Masking tape and a fine marker for labelling ropes and tubes as you cut them.

BEFORE YOU START

☐	Read the whole manual once, including Appendix D (open items). Decide items 1–5 before the stage they affect.
☐	Decide the box size: the modelled box (this manual) or the compact box (Appendix E). It changes the panels, three rods and three prints.
☐	Check your 5/16in rod: threaded or smooth (Open item 4).
☐	Check the hardware against Appendix A and the actual-size drawings on page 10. The M3 kit holds every screw size used.
☐	Have at least 9 m of rope, a 5 m roll of PTFE tube, 57 BBs plus spares, and the crimp sleeves.
☐	Print plates in the Stage 1 order and do each fit test as the plate comes off.

SYSTEM OVERVIEW

WHAT YOU ARE BUILDING

Three levers and a wheel on a wooden control box drive the EX-MA arm through wire ropes. Each control moves exactly one motion: **boom**, **stick**, **bucket** or **slew**. The ropes run from the box through a 2in conduit under the sandbox wall into the pedestal, up the turret's centre tube and along the arm.

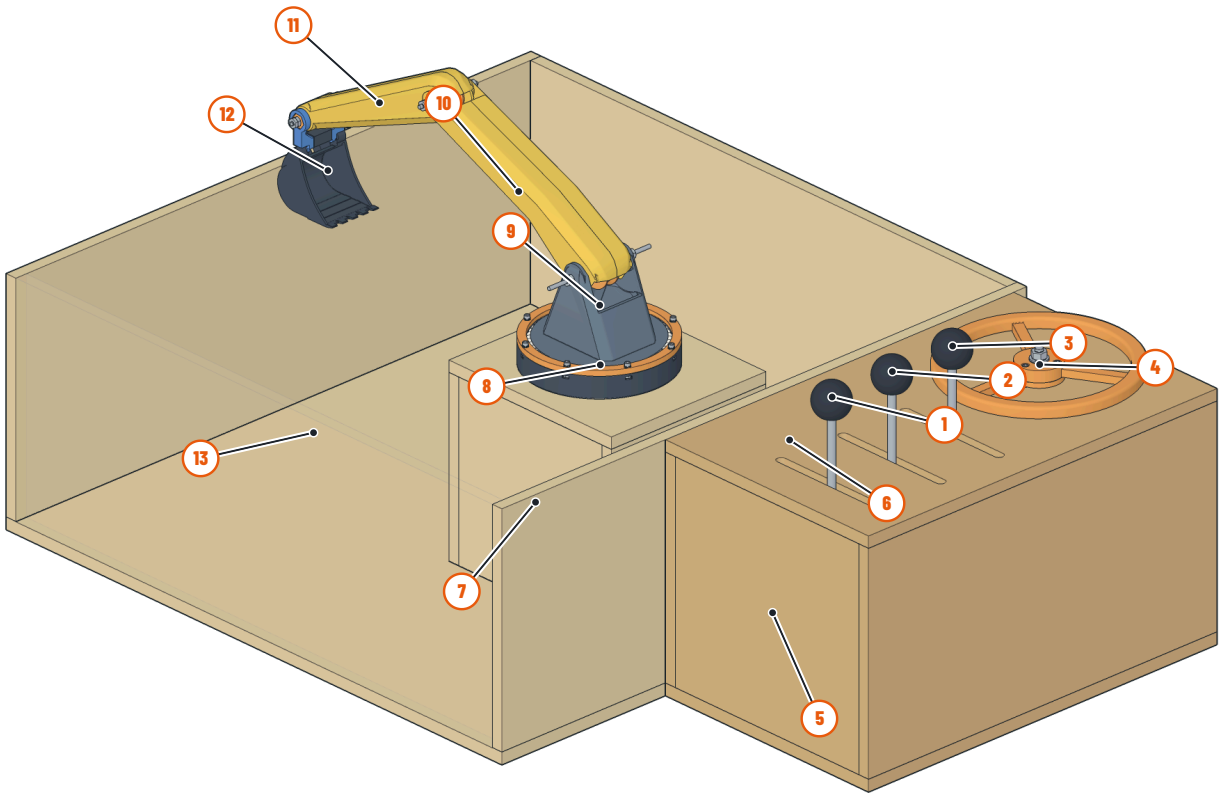


Fig. 1 The complete machine as modelled (joints at their modelled pose, the near sandbox wall see-through).

1. Boom lever
2. Stick lever
3. Bucket lever
4. Slew wheel
5. Control box (plywood)
6. Removable box top — the lever slots are the end stops
7. Pedestal (inside the sandbox)
8. Base with the printed slewing ring (57 BBs)
9. Tower (turret)
10. Boom
11. Stick
12. Bucket
13. Sandbox, 610 × 610 mm

MAIN DIMENSIONS VERIFIED

Sandbox (outside)	610 × 610 × 242 mm
Control box (outside, u × v × z)	250 × 405 × 242 mm
Pedestal (outside)	200 × 180 × 190 mm
Base / slewing ring	Ø140 mm
Boom / stick, pin to pin	209 / 175 mm
Knob tops above the floor	339 mm

The control box is the size in the CAD model. A smaller box (214 × 362 × 156) was specified on 2026-09-28 but not modelled; see Appendix E before you cut the box panels.

SYSTEM OVERVIEW

HOW IT WORKS

Every circuit is a **pull-pull loop**: one rope, anchored on two drums. Turning the control drum pulls one strand and lets out the other by the same amount, so the joint drum — and the joint — turns. Nothing else is connected to that rope, so each control moves only its own joint.

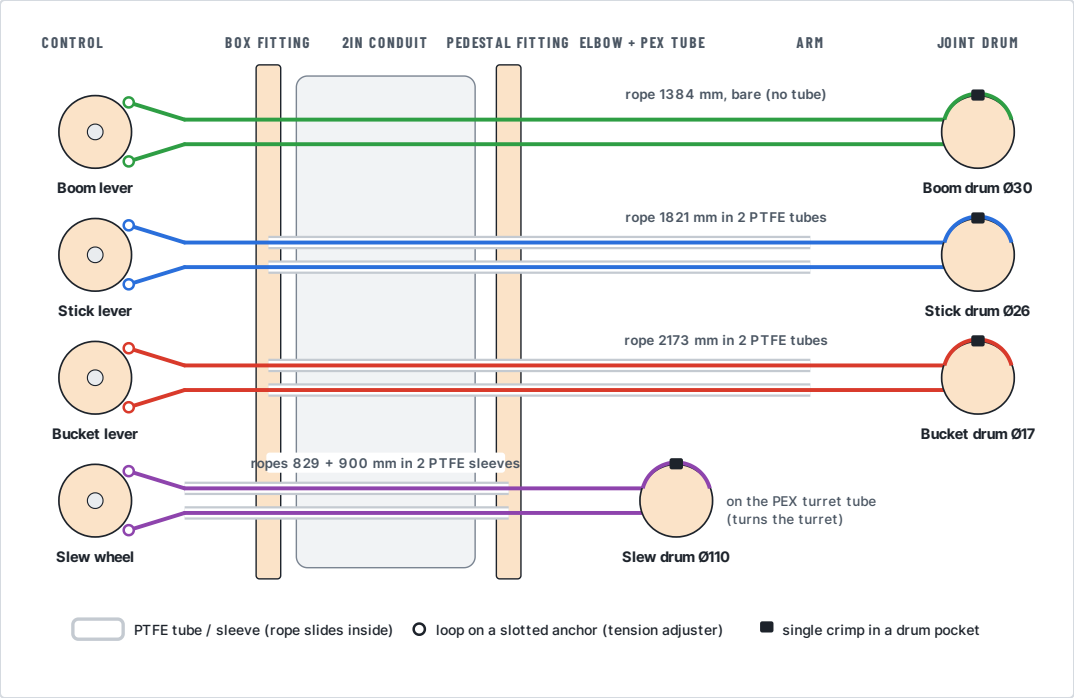


Fig. 2 Rope circuits (the “wiring diagram” of this machine). Each lever drives a Ø60 drum on the common lever axle; the slew wheel drives a Ø110 two-lane spool.

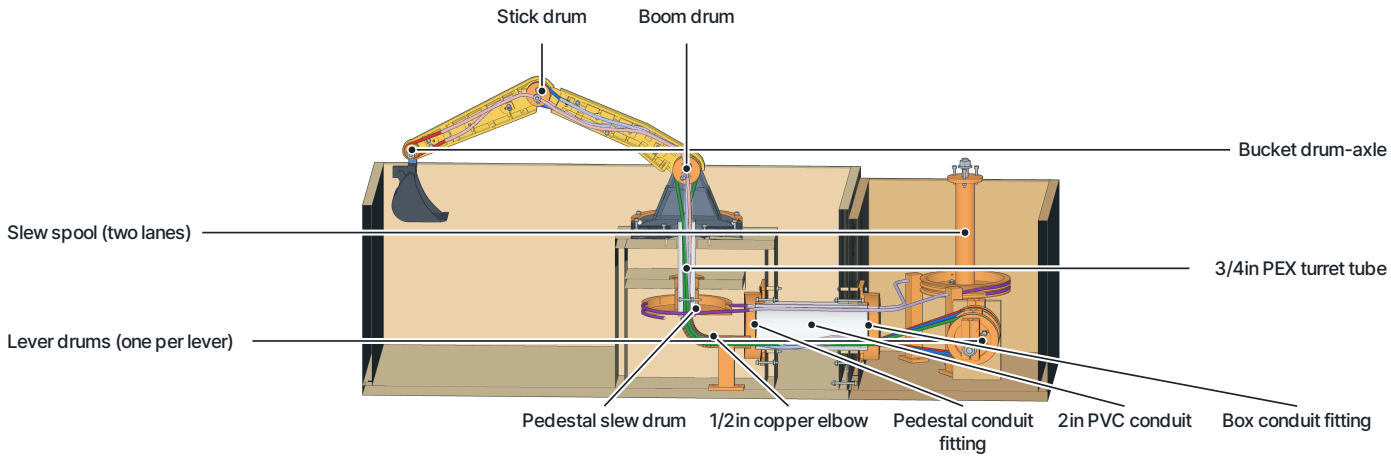


Fig. 3 Every rope as modelled, from its control to its joint (cut through the middle; built pose).

Why the stick and bucket ropes run in PTFE tubes

Their ropes cross the boom joint (and the bucket ropes the stick joint). Inside a tube that is anchored at the box fitting and at a stop in the arm, the rope length does not change when another joint moves. The tubes cross each joint through the pivot axis (the pins are split in two for this), and the small length change left is taken up as spare tube in the conduit. VERIFIED

Why the slew drum is in the pedestal

The turret turns on a printed ball ring. A 3/4in PEX tube clamped in the turret hub runs down through the base into the pedestal, where the slew drum is clamped to it; a shelf bushing just above the drum takes the ropes’ side pull. The arm ropes run up the inside of the same tube. VERIFIED

SYSTEM OVERVIEW

CONTROLS AND SPECIFICATIONS

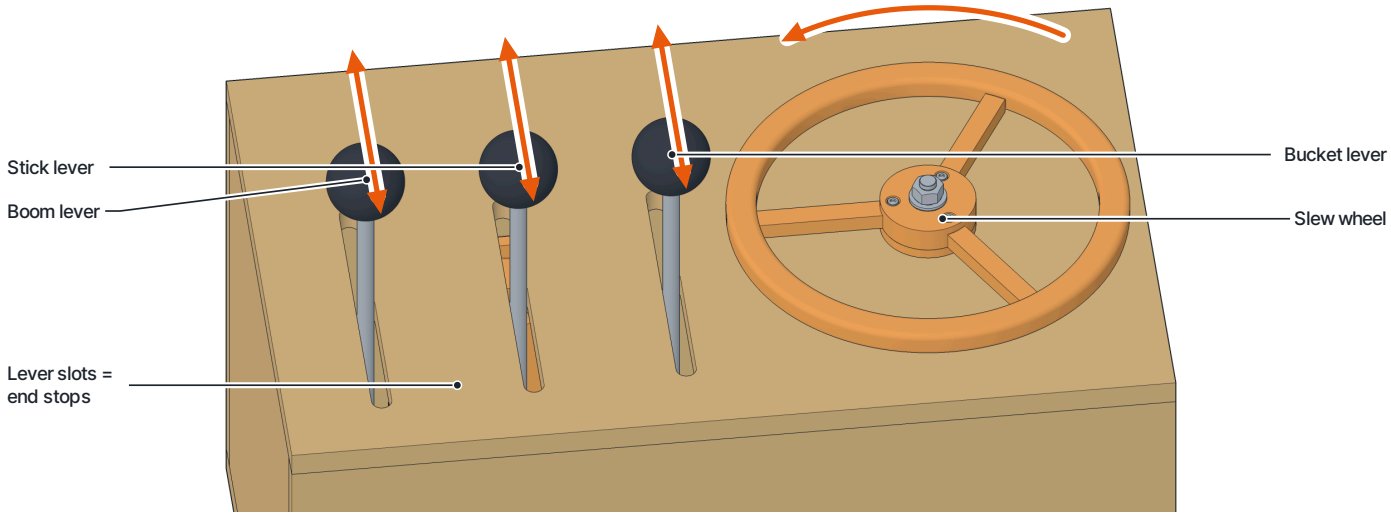


Fig. 4 The operator's view: three levers in slots, the slew wheel on the right. Orange arrows show the lever travel and the wheel.

LEVER AND WHEEL DIRECTIONS

Control	Push forward	Pull back	Joint range	Lever swing
Boom lever	boom down	boom up	-40° … +45°	42.5°
Stick lever	stick out	stick in	-45° … +50°	41.2°
Bucket lever	bucket dump	bucket curl	-55° … +85°	39.7°
Slew wheel	turn left (anticlockwise from above) → arm swings to your left; 1 : 1		−90° … +90°	180°

"Forward" = toward the sandbox (+u). Directions and ranges VERIFIED: architecture notes and kinematics report. You set the lever direction when you rig each rope (Stage 9); a reversed lever is fixed by reversing that rope's wrap on its joint drum.

KEY SPECIFICATIONS VERIFIED

Rope	1/16in galvanized 7×19
Joint drums (pitch Ø)	boom 30 · stick 26 · bucket 17
Control drums	levers Ø60 · slew spool Ø110
Slew drums	Ø110, two lanes
Slewing ring	57 × 6 mm BBs on Ø118
Joint pins	8-32 threaded rod
Axles / handles	5/16in rod
Printed parts	26 types, 29 prints, ≈1.03 kg
Materials	arm shell PETG, rest PLA+

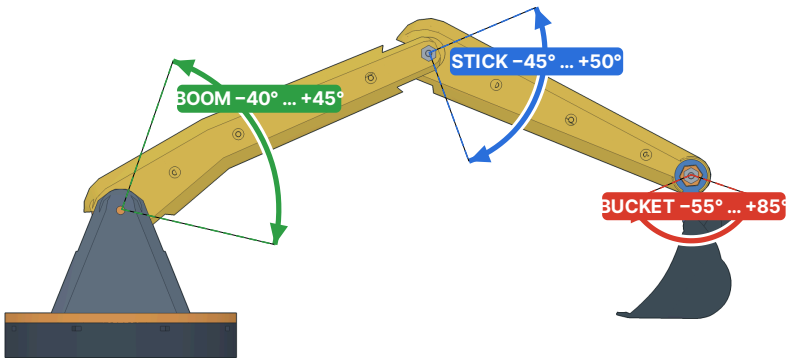


Fig. 5 What each lever moves: the working ranges set by the slots (detail in Stage 12).

PLANNING

BUILD AT A GLANCE

Twelve stages. Stages 1 and 2 are preparation; 3–8 build the machine; 9–12 rig, adjust and test it.

01 p. 14
PRINT & CHECK
29 prints on 7 plates; fit test after each plate

02 p. 15
CUT, DRILL & LABEL
plywood, rods, rope, PTFE; label every piece

03 p. 16
SANDBOX
floor + 4 walls

04 p. 17
PEDESTAL
walls, conduit fitting, copper elbow, shelf, bushing

05 p. 19
SLEWING RING & TURRET
base, PEX tube, BBs, turret, retaining ring, slew drum

06 p. 21
CONTROL BOX
carcass, lever axle + hubs, spool, posts

07 p. 24
CONDUIT, TUBES, BOX MOUNTING
conduit, 4 PTFE tubes, bolt the box on

08 p. 26
ARM
drums, ropes at the joint end, halves, pins, bucket

09 p. 30
ARM ROPES AT THE LEVERS
zero pose, loops on anchors, tension, direction

10 p. 32
SLEW ROPES
sleeves, drum pockets, spool anchors

11 p. 34
TOP, HANDLES, WHEEL
top panel, handles, knobs, wheel, drag clamps

12 p. 35
TEST & CONFIGURE
ranges, coupling, tension, friction

CHANGES FROM THE 2026-09-25 CHECKLIST DERIVED

This manual follows the order in `2026-09-25-print-and-assembly-checklist.md` with three changes the geometry forces:

- **The box is bolted on after the turret (Stage 7, not 5).** The four PTFE tubes stop in a window on the conduit side of the box fitting; they can only be seated there while that fitting is still loose. The same bolts hold the fitting, the box front and the sandbox wall together.
- **The lever axle goes in before the left box side.** With the side fitted, its nut is 2 mm from the side panel and the 205 mm rod cannot be threaded through the blocks.
- **The joint end of each arm rope is rigged while its arm half is open (Stage 8).** The crimp pockets are inside the closed arm.

OPEN ITEMS TO DECIDE BEFORE YOU START OPEN ITEM

Preparing this manual turned up the following conflicts in the design files. Read Appendix D now; items 1–3 change what you do in Stage 8.

#	Item	Affects
1	Stick-drum key screws: two M3 × 12 enter each through-hole from opposite sides and their tips overlap by 1.6 mm.	Stage 8
2	Stick joining lugs 1 and 2 at the stick root are 2.9 mm apart: their holes merge, so only one screw fits there.	Stage 8
3	Joining-lug screws cannot be reached once a boom or stick is closed: no access hole through the outer wall.	Stage 8
4	5/16in nuts on the axles need threaded rod ends; one design note calls the rod “smooth”.	Stages 6, 11
5	The cut list says the pedestal screws to the sandbox floor, but no screws are counted for it.	Stage 4
6	The 3MF plates still hold the old base and bucket ears; print the 2026-09-29 files instead.	Stage 1
7	Rope tension: the checklist says slide the anchors “outward”; the geometry says inward (toward the drum centre).	Stages 9, 10

PLANNING

TOOLS AND SUPPLIES

From the project checklist VERIFIED. Everything else is in the bill of materials (Appendix A).

HAND TOOLS

2.5 mm hex key	M3 socket head cap screws
2 mm hex key	M3 button head screws (rope anchors)
5.5 mm wrench or nut driver	M3 nuts
11/32in wrench (×2)	8-32 jam nuts — one to hold, one to turn
1/2in wrench	5/16in nuts on the axles
Screwdriver (Phillips)	#6 and #8 wood screws
Crimp tool for 1/16in sleeves	single and double crimp sleeves
Wire-rope cutter	1/16in 7×19 rope
PTFE tube cutter or new razor blade	cut square
Pipe cutter or hacksaw; deburring tool	PVC conduit, copper

POWER TOOLS AND CONSUMABLES

Drill + 3.5 mm and 2 mm bits	M3 clearance; wood-screw pilots
Drill bits for the panel holes	Ø62 (hole saw), Ø30.4, Ø26.2, Ø24.4, Ø22 counterbore, Ø8.5, Ø4.6, Ø3.0 — sizes from the cut list VERIFIED
Saw for 12 mm plywood	21 panels (Appendix B)
Wood glue	every panel joint
CA or epoxy	bucket ears to the bucket; knobs on the handles
Silicone	conduit seal at the sandbox wall
Cable tie (1)	copper elbow to its support
Masking tape + fine marker	label every rope and tube
Metal file	slot-end burrs, rod ends ASSUMPTION

3D PRINTING

The parts were laid out for a Bambu Lab A1 with a 0.6 mm nozzle; slicer projects for Bambu Studio and OrcaSlicer are in `3MF/`. Arm shell in PETG, everything else in PLA+ (SUNLU PLA+ 2.0). Stage 1 lists the settings. VERIFIED

HARDWARE BAGS BY STAGE VERIFIED

Stage	Fasteners and stock to bag
3 Sandbox	32 × #6 × 1-1/4in
4 Pedestal	40 × #6 × 1-1/4in; 5 × #6 × 3/4in; 4 × M3 × 25, 8 flat washers, 4 nuts; copper elbow; cable tie
5 Slewing ring	4 × #8 × 1in; 8 × M3 × 12; 2 × M3 × 25; 10 split-lock washers; 10 nuts; 57 BBs; PEX tube
6 Control box	32 × #6 × 1-1/4in; 11 × #6 × 3/4in; 205 + 274 mm 5/16in rod; 3 × 5/16in nut + washer
7 Conduit + box	4 × M3 × 35; 2 × M3 × 30; 12 flat washers; 6 nuts; PVC conduit; 4 arm PTFE tubes
8 Arm	8 × M3 × 12 (or M3 × 8, Open item 1); 8 × M3 × 20 + 8 nuts; 8-32 rods 2 × 42, 2 × 22, 1 × 65 mm; 12 × 8-32 nuts; 3 single crimps
9 Arm ropes	6 × M3 × 10 button head; 12 flat washers; 6 split-lock washers; 6 nuts; 6 double crimps
10 Slew ropes	2 × M3 × 10 button head; 4 flat washers; 2 split-lock washers; 2 nuts; 2 single + 2 double crimps; 2 sleeves
11 Top	16 × #6 × 1-1/4in; 9 × M3 × 20; 9 split-lock washers; 9 nuts; 3 handle rods; 1 × 5/16in nut + washer

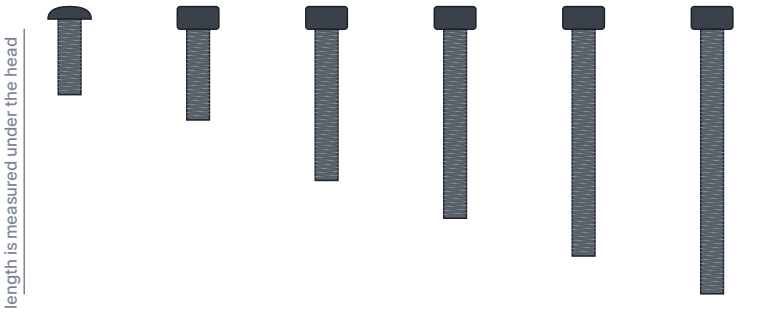
Counts follow the bill of materials and the hardware model; the M3 split-lock and flat-washer totals (27 / 36) match.

PARTS IDENTIFICATION

HARDWARE AT ACTUAL SIZE

Print this page at 100 % (no “fit to page”) and lay each screw on its drawing to identify it. The scale bar at the bottom must measure 100 mm. Sizes from the screw standards used in the hardware model VERIFIED.

M3 SCREWS — HEAD UP, ACTUAL SIZE



M3 × 10	M3 × 12	M3 × 20	M3 × 25	M3 × 30	M3 × 35
button head	socket cap	socket cap	socket cap	socket cap	socket cap
H01 · ×8	H02 · ×16	H03 · ×17	H04 · ×6	H05 · ×2	H06 · ×4

M3 NUT AND WASHERS



M3 nut	flat washer	split-lock
H07 · ×45	H08 · ×36	H09 · ×27

8-32 ROD AND NUT (HILLMAN)

22 mm · stick split pin · H11 ×2

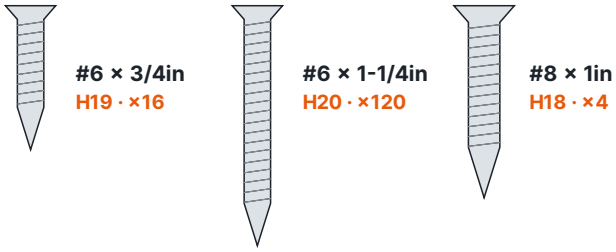
42 mm · boom split pin · H10 ×2

65 mm · bucket pin · H12 ×1



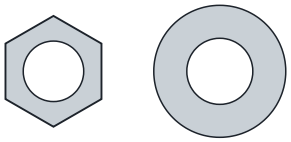
8-32 nut
H13 ×12

WOOD SCREWS (FLAT HEAD)



#6 × 3/4in	#6 × 1-1/4in	#8 × 1in
H19 · ×16	H20 · ×120	H18 · ×4

5/16in NUT AND WASHER



H17 · ×4 each (axle ends)

STOCK (ACTUAL SIZE)



6 mm BB

PTFE tube, end

rope Ø1.6

crimp (nominal)



100 mm — measure this to check your print is at 100 %

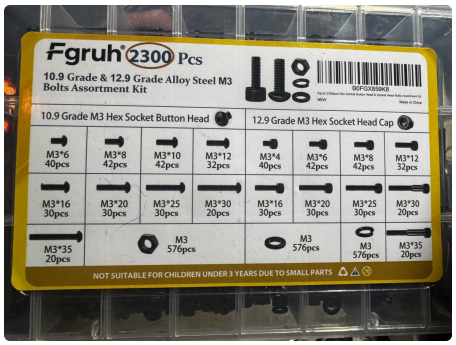


Photo Your M3 kit (Fgruh, 2300 pc): socket head caps and button heads in 4–35 mm, nuts, flat and split-lock washers.



Photo Your Hillman parts: 3 ft of 8-32 threaded rod and 22 hex nuts (used); the #4-40 and #6-32 machine screws are not needed.

M3 × 12 VS M3 × 20. They look alike in the bag. Lay them on the drawing above before each step.

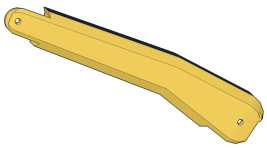
WOOD SCREWS are your own stock. Flat-head #6 and #8 are drawn; pan heads will not sit flush in the printed counterbores. ASSUMPTION

PARTS IDENTIFICATION

PRINTED PARTS

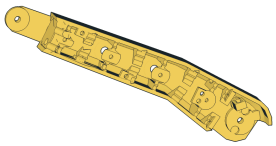
26 part types, 29 prints. Each card shows the part ID used in this manual, the quantity, the material and slicer plate, and the STL file. VERIFIED

P01 ×1



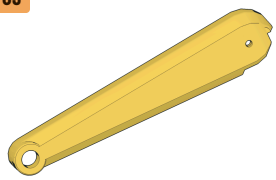
Boom half, left (+v)
PETG · plate arm 1
`STL/2026-09-24-boom-half-left.stl`

P02 ×1



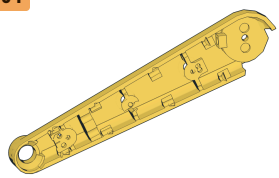
Boom half, right (−v)
PETG · plate arm 1
`STL/2026-09-24-boom-half-right.stl`

P03 ×1



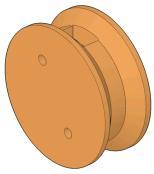
Stick half, left (+v)
PETG · plate arm 1
`STL/2026-09-24-stick-half-left.stl`

P04 ×1



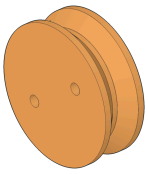
Stick half, right (−v)
PETG · plate arm 1
`STL/2026-09-24-stick-half-right.stl`

P05 ×1



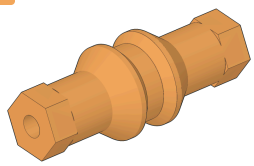
Boom drum (Ø30 pitch)
PLA+ · plate other 6
`STL/2026-09-24-boom-drum.stl`

P06 ×1



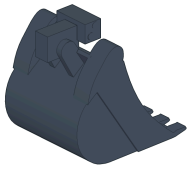
Stick drum (Ø26 pitch)
PLA+ · plate other 6
`STL/2026-09-24-stick-drum.stl`

P07 ×1



Bucket drum-axle (Ø17 pitch)
PLA+ · plate other 6
`STL/2026-09-24-bucket-drum-axle.stl`

P08 ×1



Bucket
PLA+ · plate other 6
`STL/2026-09-24-bucket.stl`

P09 ×1



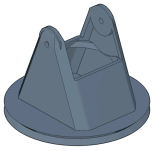
Bucket ear, left (G)
PLA+ · plate other 6 · **new file**
`STL/2026-09-29-bucket-ear-left-cad.stl`

P10 ×1



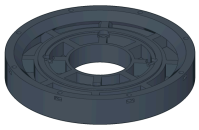
Bucket ear, right (H)
PLA+ · plate other 6 · **new file**
`STL/2026-09-29-bucket-ear-right-cad.stl`

P11 ×1



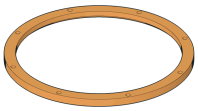
Tower (turret)
PLA+ · plate other 4
`STL/2026-09-24-tower.stl`

P12 ×1



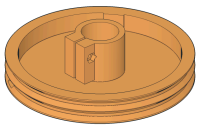
Base (lower race)
PLA+ · plate other 4 · **new file**
`STL/2026-09-29-base.stl`

P13 ×1



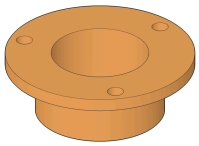
Slewing-ring retaining ring
PLA+ · plate other 5
`STL/2026-09-24-slewing-ring-retaining-ring.stl`

P14 ×1



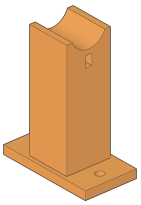
Pedestal slew drum (Ø110)
PLA+ · plate other 1
`STL/2026-09-24-slew-drum.stl`

P15 ×1



PEX bushing
PLA+ · plate other 1
`STL/2026-09-24-slew-tube-bushing.stl`

P16 ×1

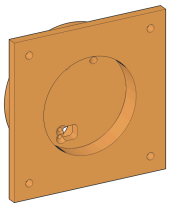


Elbow support
PLA+ · plate other 1
`STL/2026-09-24-elbow-support.stl`

PARTS IDENTIFICATION

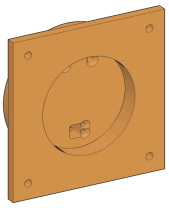
PRINTED PARTS (CONTINUED)

P17 ×1



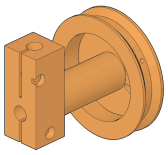
Conduit fitting, pedestal end
PLA+ · plate other 1
`STL/2026-09-24-conduit-end-fitting-pedestal.stl`

P18 ×1



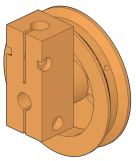
Conduit fitting, box end
PLA+ · plate other 3
`STL/2026-09-24-conduit-end-fitting-box.stl`

P19 ×1



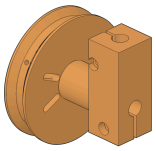
Lever hub, boom
PLA+ · plate other 2
`STL/2026-09-24-lever-hub-boom.stl`

P20 ×1



Lever hub, stick
PLA+ · plate other 2
`STL/2026-09-24-lever-hub-stick.stl`

P21 ×1



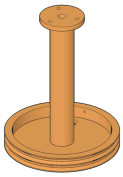
Lever hub, bucket
PLA+ · plate other 3
`STL/2026-09-24-lever-hub-bucket.stl`

P22 ×3



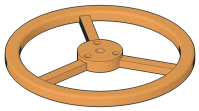
Lever knob (Ø36)
PLA+ · plate other 2
`STL/2026-09-24-lever-knob.stl`

P23 ×1



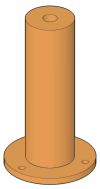
Slew spool (Ø110)
PLA+ · plate other 3
`STL/2026-09-24-slew-spool.stl`

P24 ×1



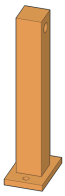
Slew wheel (Ø184)
PLA+ · plate other 2
`STL/2026-09-24-slew-wheel.stl`

P25 ×1



Spool riser
PLA+ · plate other 2
`STL/2026-09-24-spool-riser.stl`

P26 ×2



Slew sleeve post
PLA+ · plate other 2
`STL/2026-09-24-slew-tube-post.stl`

USE THE CURRENT FILES FOR P09, P10 AND P12. The base was rebuilt on 2026-09-29 (no supports needed under the race rim, the four unused outer holes filled) and both bucket ears were rebuilt as clean CAD solids (reinforced joint, no mesh defects). The slicer projects in `3MF/` still contain the older versions: replace them before printing plates 4 and 6 (Stage 1). Do not use the superseded patched-mesh files `STL/2026-09-29-bucket-ear-left.stl` / `-right.stl`.

ARM-SHELL HALVES are mirror images. The **left** halves (P01, P03) sit on the +v side, the operator's left when the arm points away from the box.

PARTS IDENTIFICATION

WOOD AND STOCK

21 panels, all 12 mm plywood (1/2in also works). Sizes from the cut list generated by `build-box.py` VERIFIED. Hole positions are drawn in Appendix B.

ID	Panel	Size (mm)	Holes	Note
W01	**Control box** · box floor	405 × 250	1	
W02	**Control box** · box front	405 × 218	7	
W03	**Control box** · box rear	405 × 218	—	
W04	**Control box** · box right side	226 × 218	—	
W05	**Control box** · box left side	226 × 218	—	
W06	**Control box** · box top (removable)	405 × 250	4	screwed to the sides, front and rear; lift off for service
W07	**Control box** · axle bearing block left	100 × 60	1	screwed to the floor from below (2 x #6)
W08	**Control box** · axle bearing block right	100 × 60	1	screwed to the floor from below (2 x #6)
W09	**Pedestal** · pedestal top	200 × 180	5	
W10	**Pedestal** · pedestal wall -u (conduit)	180 × 178	5	
W11	**Pedestal** · pedestal wall +u (removable access)	180 × 178	—	screwed on; remove to reach the slew drum clamp and the elbow
W12	**Pedestal** · pedestal wall -v	178 × 176	—	
W13	**Pedestal** · pedestal wall +v	178 × 176	—	
W14	**Pedestal** · pedestal shelf	156 × 148	1	rests on the two cleats; carries the PEX bushing just above the slew drum
W15	**Pedestal** · pedestal shelf cleat -v	148 × 12	—	
W16	**Pedestal** · pedestal shelf cleat +v	148 × 12	—	
W17	**Sandbox** · sandbox floor	610 × 610	—	the pedestal and the elbow support screw to it
W18	**Sandbox** · sandbox wall -u (box side)	610 × 230	7	
W19	**Sandbox** · sandbox wall +u	610 × 230	—	
W20	**Sandbox** · sandbox wall -v	586 × 230	—	
W21	**Sandbox** · sandbox wall +v	586 × 230	—	

STOCK ITEMS

ID	Item	Qty / length	Used for
S01	1/16in galvanized 7×19 wire rope	5 pieces	1384 / 1821 / 2173 / 829 / 900 mm
S02	1/16in single crimp sleeve	5	3 joint-drum midpoints + 2 slew rope ends
S03	1/16in double crimp sleeve	8	6 arm tail loops + 2 slew loops
S04	PTFE tube 4 mm OD × 2 mm ID	6 pieces	647 / 658 / 813 / 822 / 339 / 407 mm
S05	6 mm airsoft BB (or 1/4in steel ball)	57	slewing ring (+ about 5 spare)
S06	3/4in PEX-B tube	120 mm	turret tube (turns with the turret)
S07	1/2in copper 90° sweep elbow + 50 mm pipe	1	fixed elbow in the pedestal; deburr both ends
S08	2in sch 40 PVC pipe	141 mm	conduit between box and pedestal
S09	Wood glue; CA or epoxy; silicone; cable tie	—	panel joints; ears + knobs; conduit seal; elbow

STAGE 1

PRINT AND CHECK THE PARTS

Seven plates in build order; a fit test after each plate catches a bad print before it is buried in the assembly.

CONFIGURATION: SLICER SETTINGS VERIFIED

STORED IN THE PROJECT FILES — OPEN, CHECK, PRINT

Printer preset	**A1 0.6 Fast Start (your preset)**
Process	**0.30mm Strength @BBL A1 0.6 nozzle — 0.3 mm layers, 4 walls**
Infill	**25 % (40 % on drums, hubs, clamps and pinned parts)**
Supports	**tree, only where the plate marks them**
PETG filament	**Generic PETG @BBL A1 — arm shell only**
PLA+ filament	**SUNLU PLA+ 2.0 @BBL A1 — everything else**
PLA+ speed (project)	**flow cap 28 mm³/s at 230 °C; inner walls / infill / supports 200 mm/s; accel 10000**
If infill looks thin	**lower the filament's max volumetric speed to 25**

Fig. 6 Summary of the settings saved in `3MF/2026-09-27-{bambu,orca}-*.3mf` (not a screenshot: your slicer shows them under the printer, process and filament presets).

1. Open the project for your slicer from `3MF/`: `2026-09-27-bambu-…` for Bambu Studio 2.8.2.61 or `2026-09-27-orca-…` for OrcaSlicer. Each opens with its plates named and the settings in place.
2. In the PLA+ project, replace the base on plate 4 with `STL/2026-09-29-base.stl` and both ears on plate 6 with `STL/2026-09-29-bucket-ear-{left,right}-cad.stl`. Keep their orientation: base flat side down; ears on their outer side.
3. Every part is already in its print orientation and every wall was checked for the 0.6 mm nozzle — do not rotate or rescale parts.
4. Print the plates in the order of the table below. After each plate do its fit test (last column) before printing the next.

Material	Plate	Name	Parts	Fit test after printing
PETG	arm 1	Arm shell	P02, P01, P04, P03	bucket drum-axle journals turn freely in both stick-half nose holes
PLA+	other 1	Pedestal	P17, P14, P15, P16	copper elbow pushes into P17's socket; a PTFE tube slides through its window; the PEX tube turns in P15; P14 slides onto the PEX tube (clamp open)
PLA+	other 2	Control box 1/2	P24, P20, P19, P25, P22 ×3, P26 ×2	5/16in rod slides through each hub and into its handle socket; rod enters each knob; riser sits flat
PLA+	other 3	Control box 2/2	P18, P23, P21	rod through the hub; spool turns on the rod; 4 PTFE tubes seat together in P18's window and stop on its floor
PLA+	other 4	Turret + slewing ring 1/2	P12 (new file), P11	PEX tube slides into the tower's centre bore
PLA+	other 5	Turret + slewing ring 2/2	P13	a BB rolls freely in the race between the base rim and the ring
PLA+	other 6	Arm mechanism + bucket	P08, P05, P06, P10 + P09 (new files), P07	each drum sits flat

✓ **CHECK AFTER EVERY PLATE.** Remove all support, then do the fit test. A hub that will not slide on the rod, or a drum that rocks on the table, gets reprinted now — not after it is rigged.

ⓘ **NEW BASE (P12).** Only the eight nut-slot roofs need support (about 2 cm²; the race rim has its own 45° support built in). The bill of materials still lists 28 cm² for the old base. VERIFIED

✕ **EARS SWAPPED.** P09 (left) and P10 (right) are mirror images. Mark them L and R as soon as they come off the plate.

STAGE 2

CUT, DRILL AND LABEL THE STOCK

Plywood, rods, rope and PTFE tube. Label every piece the moment it is cut.

WIRE ROPE (1/16IN 7×19) VERIFIED

Piece	Cut (mm)	Ends
Boom circuit	1384	single crimp at the midpoint (692 mm), loop at each end
Stick circuit	1821	single crimp at the midpoint (910.5 mm), loop at each end
Bucket circuit	2173	single crimp at the midpoint (1086.5 mm), loop at each end
Slew rope A	829	single crimp at one end (pedestal drum), loop at the other (spool)
Slew rope B	900	single crimp at one end (pedestal drum), loop at the other (spool)

Lengths include 60 mm for each double-crimp loop. Total 7.1 m — have at least 9 m.

PTFE TUBE 4 × 2 MM VERIFIED

Piece	Cut (mm)	Runs
stick_hi	647	box fitting → stop in the boom tip
stick_lo	658	box fitting → stop in the boom tip
bucket_hi	813	box fitting → stop in the stick nose
bucket_lo	822	box fitting → stop in the stick nose
slew sleeve A	339	post A → pedestal fitting
slew sleeve B	407	post B → pedestal fitting

Each includes 10 mm to trim at fitting. Total 3.7 m — one 5 m roll.

RODS AND PIPE VERIFIED

Stock	Cut	For
5/16in rod	205 mm	lever axle (H14)
5/16in rod	274 mm	slew spool axle (H15)
5/16in rod	3 × 230 mm	lever handles (H16)
8-32 threaded rod	2 × 42 mm	boom split pins (H10)
8-32 threaded rod	2 × 22 mm	stick split pins (H11)
8-32 threaded rod	1 × 65 mm	bucket pin (H12)
3/4in PEX-B	120 mm	turret tube (S06)
2in sch 40 PVC	141 mm	conduit (S08), socket floor to socket floor
1/2in copper	elbow + 50 mm	pedestal elbow (S07)

1. Cut all ten rope and PTFE pieces. Wrap tape round the rope at the cut mark and cut through the tape.
2. Write the piece name on a tape flag at **both** ends of every rope and tube (e.g. “STICK — box end”).
3. On the three arm ropes, mark the midpoint with a second colour (692 / 910.5 / 1086.5 mm).
4. Cut the rods square; chamfer the ends lightly with a file so they slide through printed bores. Run a nut on and off each 8-32 piece to clean the thread.
5. Cut the PVC and copper square and deburr inside and out — a burr will cut a rope.

ROPE ENDS. Tape every cut end until it is crimped. Loose wires go through skin easily.

PLYWOOD

Cut the 21 panels to the sizes on page “Wood and stock” and drill the holes in Appendix B. Holes per panel: box floor 1 (+ Ø22 × 7 counterbore from below), box front 7, box top 1 hole + 3 slots, bearing blocks 1 each, pedestal top 5, pedestal wall –u 5, shelf 1, sandbox wall –u 7. VERIFIED

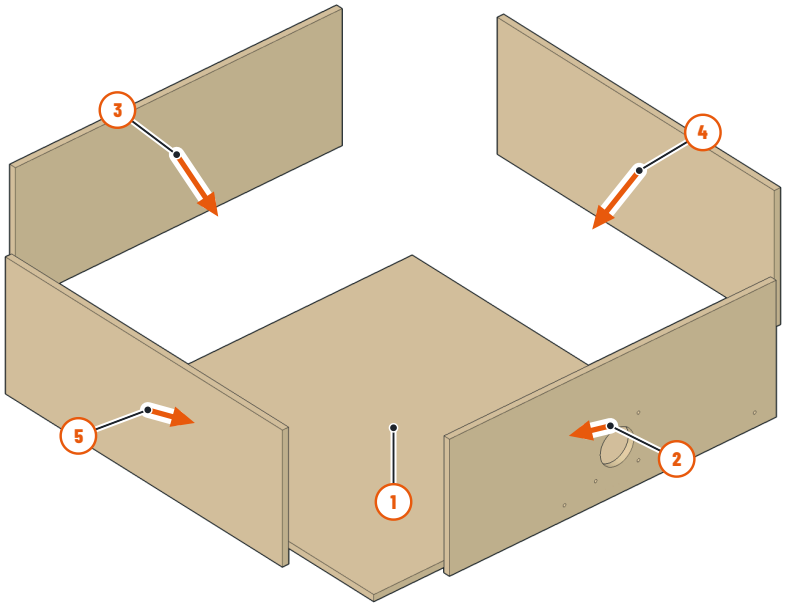
DRILL THE BOX FRONT AND THE SANDBOX WALL –U TOGETHER. Their seven holes must line up (the same M3 bolts pass through both). Clamp the two panels face to face, box front on top, and drill through both. ASSUMPTION

OPEN ITEM 4 — AXLE NUTS. The axle and spool rod ends carry 5/16in nuts. If your 5/16in rod is smooth (one design note says so), thread about 15 mm at each end of the 205 mm and 274 mm rods with a 5/16-18 die, or substitute 5/16in shaft collars. See Appendix D.

STAGE 3

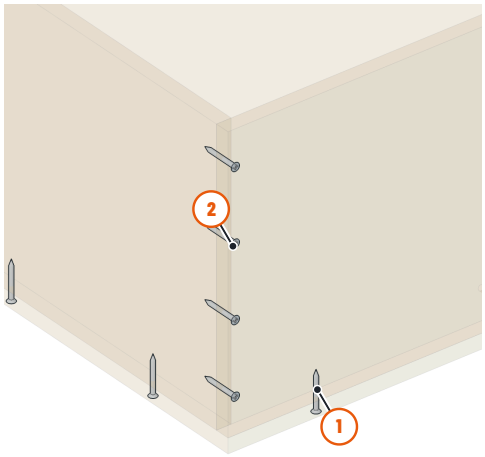
BUILD THE SANDBOX

Floor and four walls, glued and screwed. The –u wall is the drilled one: the control box bolts to its outside.



1. Sandbox floor 610 × 610
2. Wall –u (box side, drilled)
3. Wall +u
4. Wall –v
5. Wall +v

Fig. 7 Sandbox: four walls on the floor.



1. #6 × 1-1/4in up through the floor
2. #6 × 1-1/4in through wall –u into wall +v

Fig. 8 Corner joint, walls see-through.

YOU NEED

W17	Sandbox floor 610 × 610	1
W18	Wall –u (box side, 7 holes)	1
W19	Wall +u	1
W20	Wall –v	1
W21	Wall +v	1
H20	#6 × 1-1/4in wood screw	32
	Wood glue	

1. Glue and screw the four walls **on top of** the floor: 4 screws up through the floor into each wall's bottom edge.
2. The –v and +v walls fit between the –u and +u walls: 4 screws through each ±u wall into the ends of the ±v walls.
3. Put the –u wall's holes at the bottom: the Ø62 conduit hole centre is 80 mm above the wall's bottom edge and 297 mm from its –v end. VERIFIED
4. Check the box is square (equal diagonals) before the glue sets.

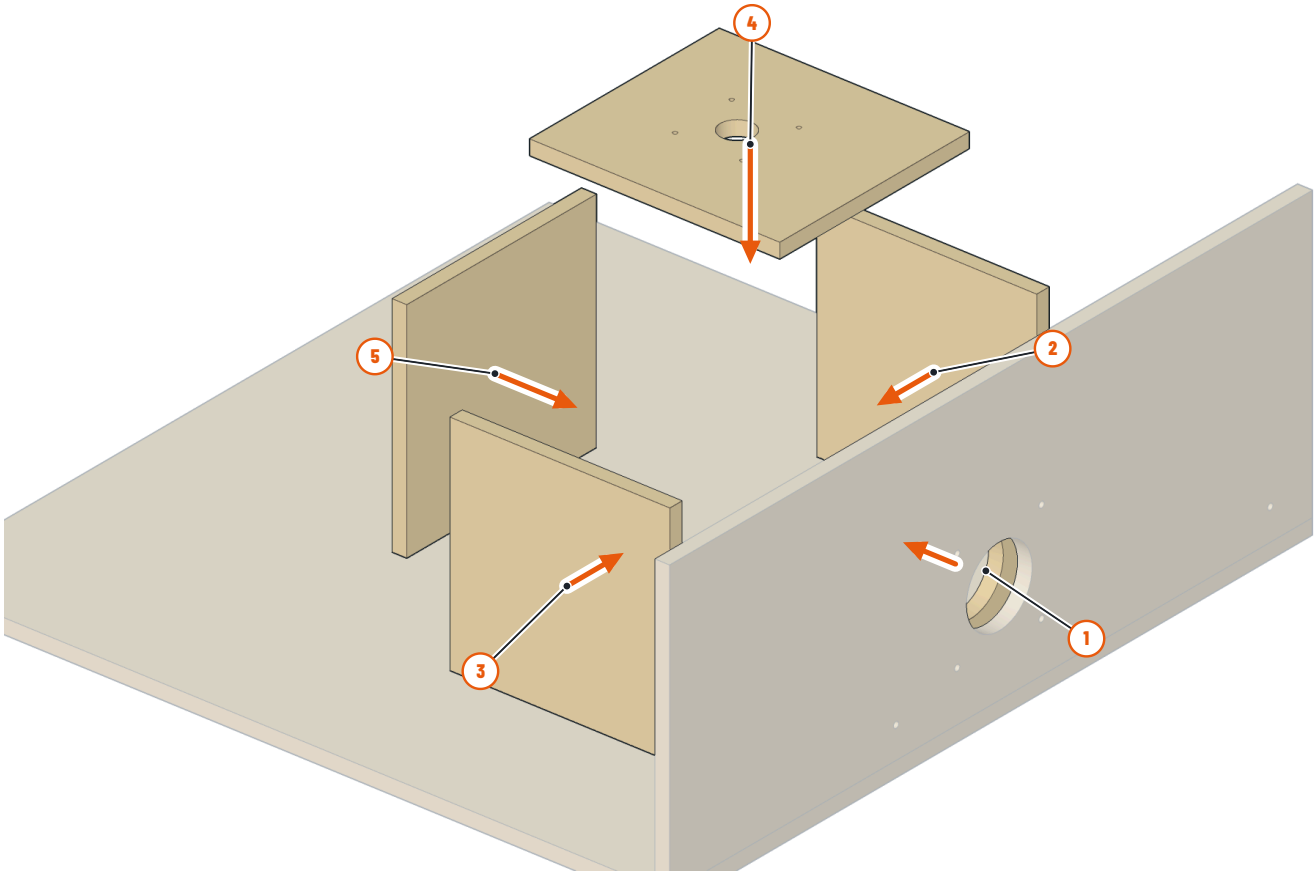
✓ **CHECK.** Walls flush with the floor edges; diagonals equal within 2 mm; the –u wall's Ø62 hole is low, near the middle.

✕ **HOLE SIDE.** Fitting the –u wall upside down puts the conduit hole 150 mm too high. The hole is 80 mm from the **bottom**.

STAGE 4

BUILD THE PEDESTAL

A plywood column inside the sandbox that carries the slewing ring on top and the slew drum inside.



1 Wall −u (conduit wall)
2 Wall −v
3 Wall +v
4 Pedestal top
5 Wall +u (removable — screw only)

Fig. 9 Pedestal walls and top on the sandbox floor. Wall +u is screwed only: it is the service hatch.

YOU NEED

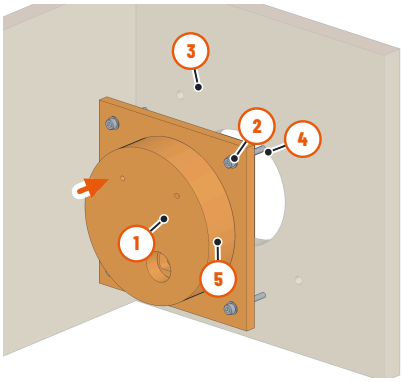
W09	Pedestal top (5 holes)	1
W10	Wall −u (conduit wall, 5 holes)	1
W11	Wall +u (removable)	1
W12	Wall −v	1
W13	Wall +v	1
H20	#6 × 1-1/4in wood screw	32

OPEN ITEM 5 — FIXING THE PEDESTAL DOWN. The cut list says the pedestal screws to the sandbox floor, but the bill of materials counts no screws for it. Suggested: 4 extra #6 × 1-1/4in up through the sandbox floor into the ±v walls (2 each). ASSUMPTION

1. Glue and screw walls −v and +v between walls −u and +u (4 screws per joint, through the ±u walls). Fit wall +u **with screws only, no glue**.
2. Glue and screw the pedestal top onto the four walls (4 screws per wall). The top's Ø26.2 hole is the PEX clearance; its four Ø3 holes are pilots for the base screws.
3. Stand the pedestal on the sandbox floor, centred side to side, with wall −u (the drilled one) facing the sandbox's −u wall.
4. Set the gap between the inner face of the sandbox −u wall and the outer face of the pedestal's wall −u to **81 mm**, so the two Ø62 conduit holes line up. VERIFIED

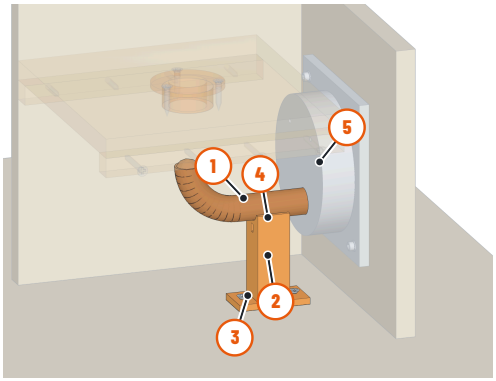
✓ **CHECK.** Sight through the sandbox wall's Ø62 hole: you see the pedestal wall's Ø62 hole centred behind it. The pedestal top is level.

STAGE 4 — CONTINUED: CONDUIT FITTING, COPPER ELBOW, SHELF



1. Pedestal conduit fitting (P17)
2. 4 × M3 × 25 + flat washer
3. Flat washer + M3 nut outside
4. Wall –u
5. Copper socket (elbow)

Fig. 10 P17 on wall –u, seen from inside the pedestal.



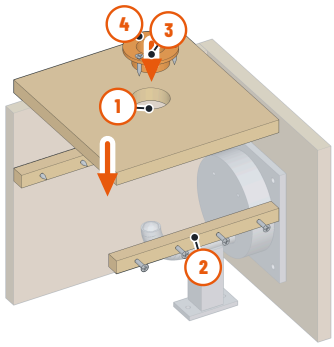
1. 1/2in copper elbow (pushed into the fitting socket)
2. Elbow support (P16)
3. 2 × #6 × 3/4in into the sandbox floor
4. Cable-tie slot
5. Pedestal fitting

Fig. 11 Copper elbow and its support.

YOU NEED (THIS PAGE)

P17	Conduit fitting, pedestal end	1
H04	M3 × 25 socket cap	4
H08	M3 flat washer	8
H07	M3 nut	4
S07	1/2in copper elbow + 50 mm pipe	1
P16	Elbow support	1
H19	#6 × 3/4in wood screw	5
	Cable tie	1
W14	Shelf	1
W15	Shelf cleats	2
H20	#6 × 1-1/4in wood screw	8
P15	PEX bushing	1

1. Remove wall +u. Hold P17's square flange flat against the **inside** of wall –u: the conduit socket opens toward the wall's Ø62 hole; the round boss, with the small copper socket in its end face, points into the pedestal. Line up the 4 corner holes.
2. Bolt it: M3 × 25 + flat washer from inside (head on the fitting flange), flat washer + nut outside the wall. Tighten evenly.
3. Push the copper elbow's horizontal leg into the fitting's copper socket (8 mm deep); the other leg points straight up.
4. Set the elbow support under the horizontal leg, screw it to the sandbox floor (2 × #6 × 3/4in) and cable-tie the elbow into its saddle.
5. Screw the two cleats to the inside of walls –v and +v (#6 × 1-1/4in), lay the shelf on them, and screw the PEX bushing into the shelf hole, flange on top (3 × #6 × 3/4in).



1. Shelf (rests on the cleats)
2. Shelf cleats (screwed to walls ±v)
3. PEX bushing (P15), flange up
4. 3 × #6 × 3/4in

Fig. 12 Cleats, shelf and PEX bushing (shelf and bushing lifted out).

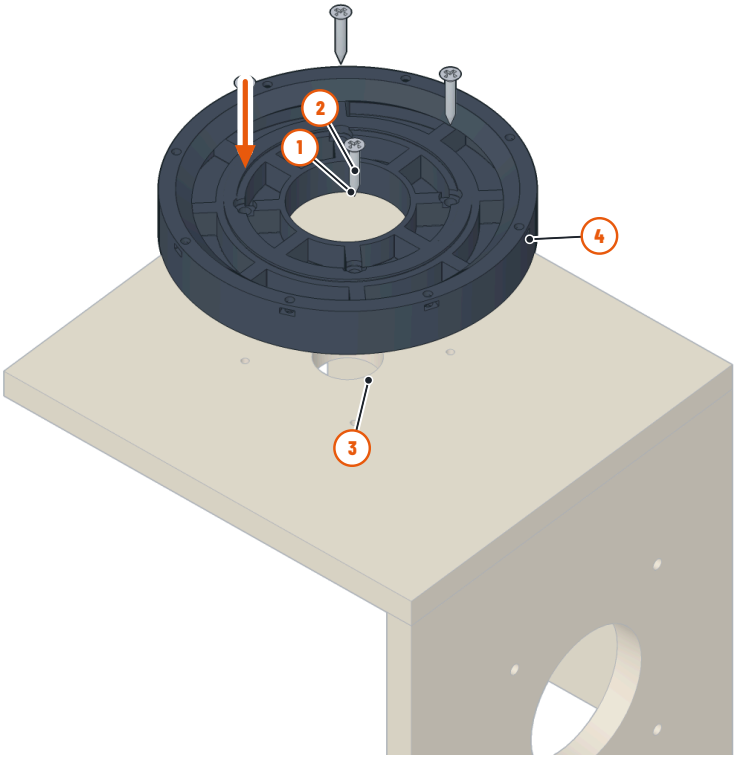
✓ **CHECK.** The elbow's top opening is directly under the bushing's bore (sight down through the bushing). The copper does not move when you pull it. **VERIFIED**

✗ **FITTING TURNED ROUND.** If the big socket faces into the pedestal, the conduit cannot reach it and the copper socket faces the wall.

STAGE 5

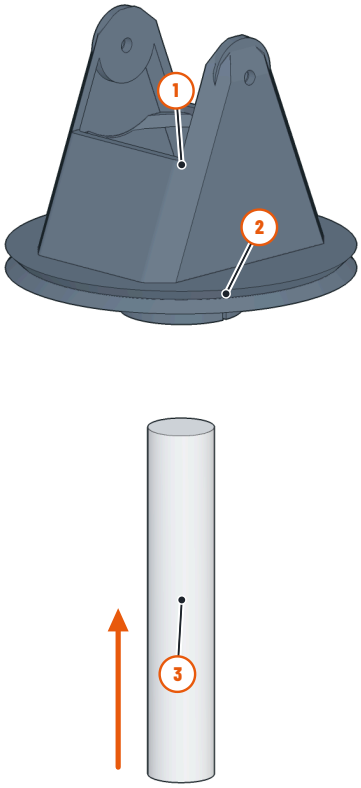
SLEWING RING AND TURRET

The base screws to the pedestal; the turret rides on 57 BBs and is held down by the retaining ring.



1. Base (P12) — current file 2026-09-29-base
2. 4 × #8 × 1in wood screw in the counterbores
3. Pedestal top (Ø3 pilots on r 38)
4. Nut slots in the rim (8)

Fig. 13 Base onto the pedestal top.



1. Tower (P11)
2. Turret hub clamp: M3 × 25 + lock washer + nut
3. 3/4in PEX-B tube, 120 mm

Fig. 14 PEX tube into the turret hub.

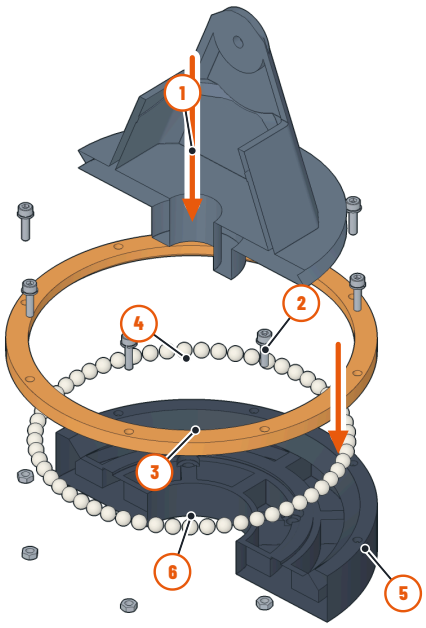
YOU NEED

P12	Base (2026-09-29 file)	1
H18	#8 × 1in wood screw	4
P11	Tower	1
S06	3/4in PEX-B tube, 120 mm	1
H04	M3 × 25 (hub clamp)	1
H09	M3 split-lock washer	1
H07	M3 nut	1

1. Set the base on the pedestal top with its centre bore over the Ø26.2 hole. Turn it until its four counterbored holes (45° positions, r 38 mm) sit over the four Ø3 pilots.
2. Screw it down with 4 × #8 × 1in in the counterbores. Do not over-tighten into the plywood.
3. Push the PEX tube up into the tower's clamp hub from below until its top is **3 mm below the top face of the turret flange** (tube top at z 22, flange top at z 25). VERIFIED
4. Tighten the hub's pinch clamp: M3 × 25 through the hub, split-lock washer and nut on the far side. The tube must not turn in the hub.

! **CLAMP THE TUBE BEFORE THE TURRET GOES ON.** Once the turret sits on the balls, the hub clamp is inside the base and cannot be reached. VERIFIED

STAGE 5 — CONTINUED: BALLS, RETAINING RING, SLEW DRUM

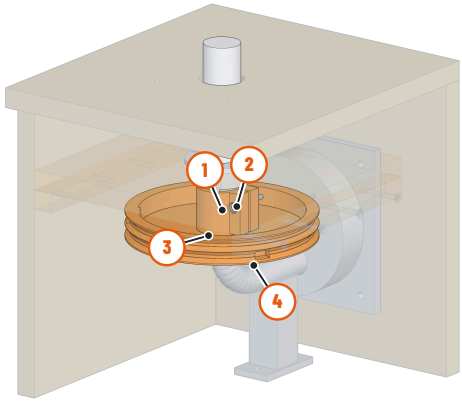


1. Turret (tower + PEX tube) lowered last
2. 8 × M3 × 12 + split-lock washer
3. Retaining ring (P13)
4. 57 × 6 mm BBs in the race
5. 8 × M3 nut, slid into the rim slots
6. Base (P12)

Fig. 15 Slewing ring, exploded and cut in half to show the race.

YOU NEED (THIS PAGE)

S05	6 mm BBs	57
P13	Retaining ring	1
H02	M3 × 12 socket cap	8
H09	M3 split-lock washer	9
H07	M3 nut	9
P14	Pedestal slew drum	1
H04	M3 × 25 (drum clamp)	1



1. Slew drum (P14) on the PEX tube
2. Pinch clamp: M3 × 25 + lock washer + nut
3. PEX tube end 2 mm below the drum
4. Copper elbow end inside the PEX bore

Fig. 16 Slew drum on the PEX tube inside the pedestal (wall +u removed).

5. Slide an M3 nut into each of the 8 slots in the outside of the base rim, flat side up, until it lines up with the hole above it.
6. Lay the 57 BBs in the base's race (the 45° groove round the rim). They fill the circle with about 0.4 mm between balls.
7. From inside the pedestal (wall +u off), hold the slew drum under the shelf, hub up. Lower the turret: the PEX tube goes down through the base, the pedestal top, the bushing and into the drum. The turret flange settles on the balls.
8. Lay the retaining ring on the rim and fit the 8 × M3 × 12, a split-lock washer under each head, into the nuts. Tighten them in a cross pattern, a little at a time.
9. Slide the drum up or down the tube until the tube's end is 2 mm below the drum's underside, and tighten its pinch clamp (M3 × 25, lock washer, nut) lightly. Its rotation is set in Stage 10.

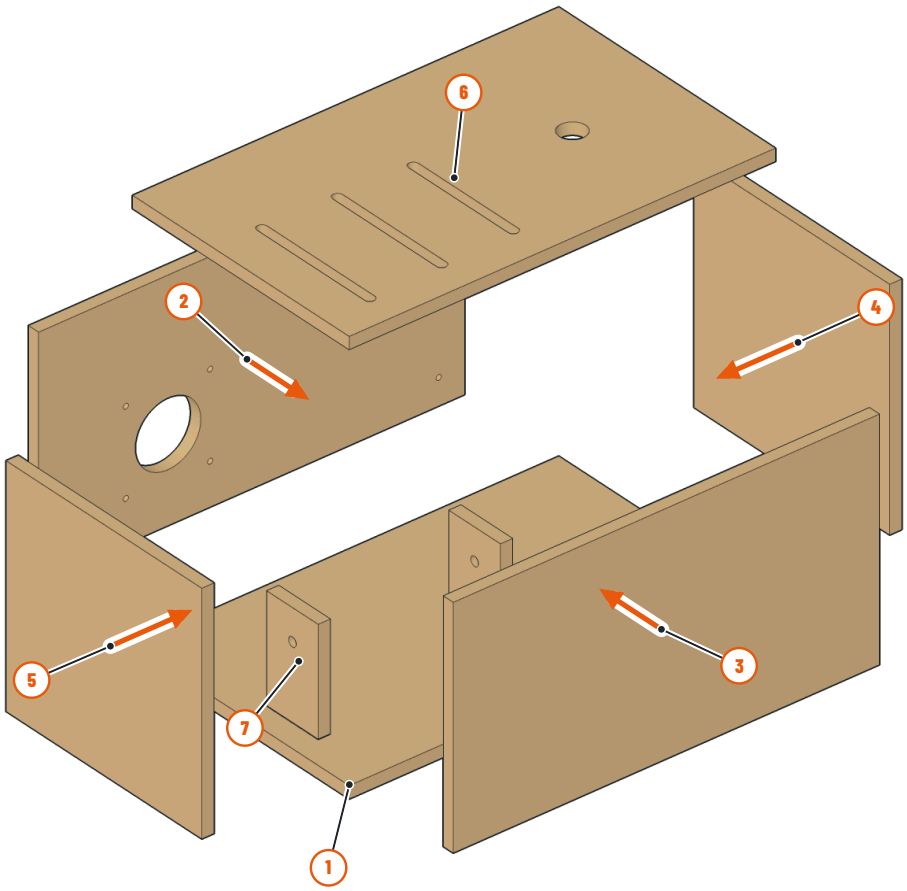
✓ **CHECK — TURRET.** The turret turns by hand with no rocking and no lift. The model allows 0.65 mm lift and 1.1° rock before parts touch. If it rocks, shim under the retaining ring; if it binds, loosen the ring screws evenly a quarter turn. VERIFIED

! **COUNT THE BALLS.** 57 balls. One missing makes a gap that knocks as the turret turns; one extra jams the ring.

STAGE 6

BUILD THE CONTROL BOX

Box carcass, lever axle with its three hubs, the slew spool and the two sleeve posts. The left side and the top go on later.



1 Floor
2 Front (to the sandbox; drilled)
3 Rear
4 Right side
5 Left side (fit after the lever axle)
6 Top (removable, fitted last)
7 Bearing blocks

Fig. 17 Control box panels.

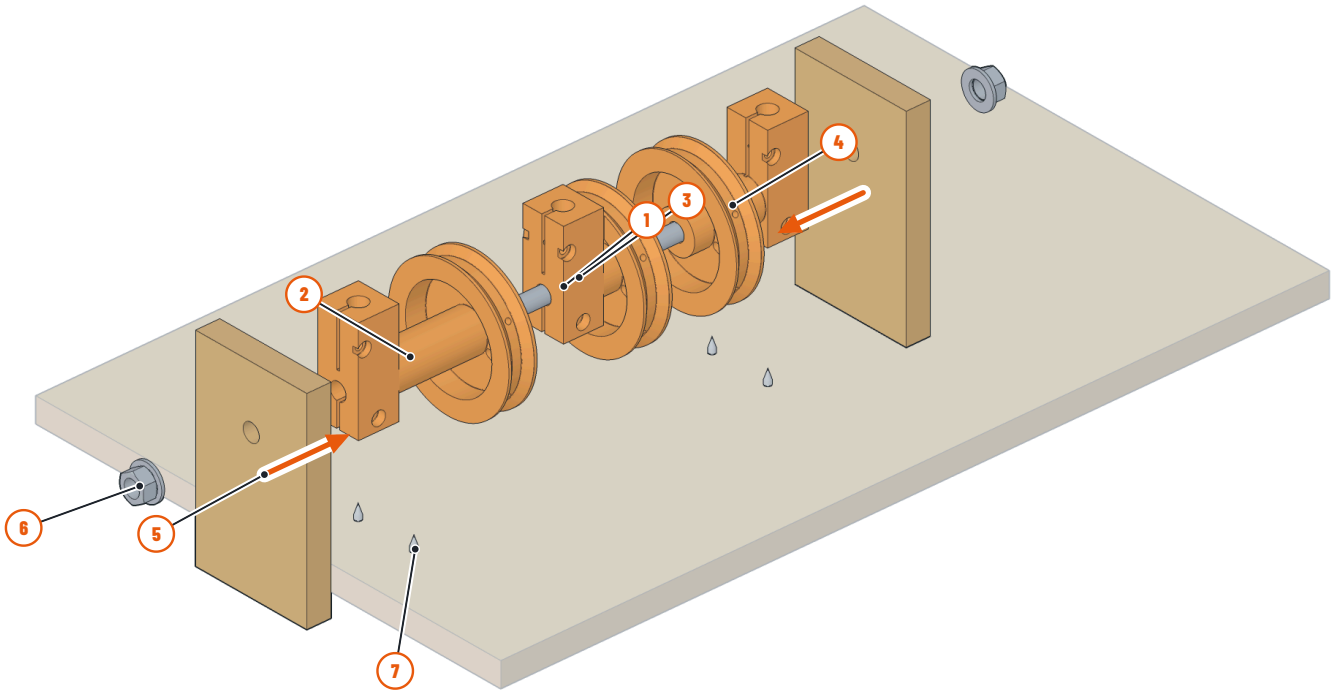
YOU NEED

W01	Box floor	1
W02	Box front (7 holes)	1
W03	Box rear	1
W04	Box right side	1
W05	Box left side (later in this stage)	1
H20	#6 × 1-1/4in wood screw	32
	Wood glue	

1. Glue and screw the front, rear and right side onto the floor (4 screws per edge, up through the floor).
2. Join the front and rear to the right side (4 screws per joint, through the front and rear).
3. The front's Ø62 hole is low (centre 80 mm above the floor's top face) and right of centre: 282 mm from the box's right-hand (−v) side, 123 mm from its left. VERIFIED
4. Leave the left side and the top off for now.

! **BOX SIZE.** These are the modelled box sizes (250 × 405 × 242). If you want the smaller box of Appendix E, its panels, rods and two reprinted hubs all change — decide before cutting.

STAGE 6 — CONTINUED: LEVER AXLE AND HUBS



1. 5/16in lever axle, 205 mm
2. Boom hub (P19) — operator's left
3. Stick hub (P20) — centre
4. Bucket hub (P21) — right
5. Bearing block (W07/W08)
6. 5/16in SAE washer + nut, each end
7. 2 × #6 × 3/4in from below, each block

Fig. 18 Lever axle stack, exploded along the axle: blocks, three hubs, washers and nuts.

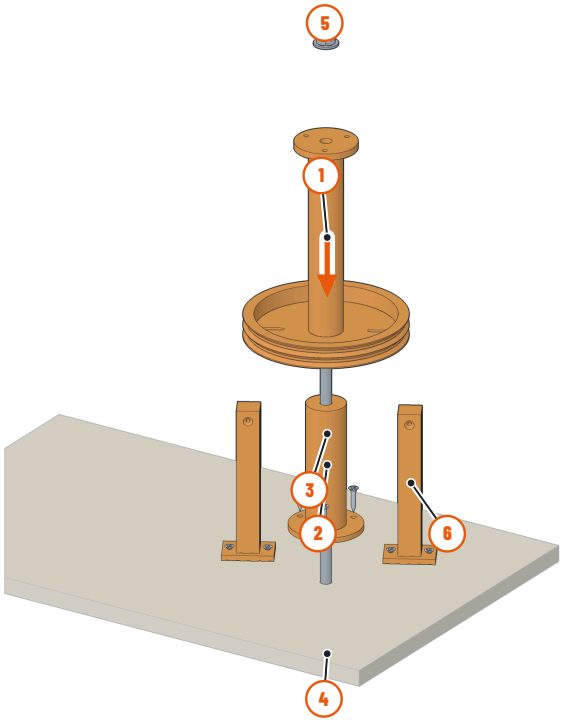
YOU NEED (THIS PAGE)

W07	Axle bearing block left	1
W08	Axle bearing block right	1
H14	5/16in rod, 205 mm	1
P19	Lever hub, boom	1
P20	Lever hub, stick	1
P21	Lever hub, bucket	1
H17	5/16in nut + SAE washer	2
H19	#6 × 3/4in wood screw	4

1. On the bench, slide onto the rod in this order from the left (+v) end: left block, **boom** hub, **stick** hub, **bucket** hub, right block. The handle sockets all point the same way (up).
2. Centre the rod (102.5 mm each side of the middle). Fit a washer and a nut outside each block, finger-tight plus a quarter turn: the hubs must still turn freely. They are 0.5 mm apart by design. VERIFIED
3. Stand the stack on the box floor: axle 93 mm from the floor's rear (−u) edge, left block's outer face 22.5 mm in from the floor's left (+v) edge. Screw each block from below: 2 × #6 × 3/4in. VERIFIED
4. Now fit the left side (glue + 4 screws per edge).

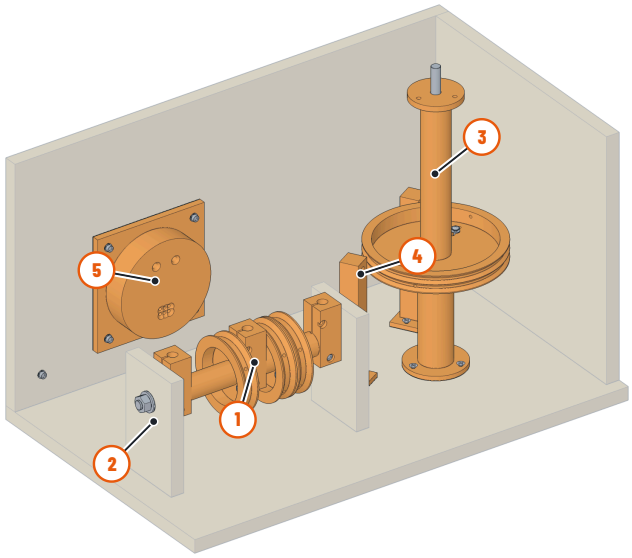
HUB ORDER. Boom (P19) on the operator's left, stick (P20) centre, bucket (P21) right. The hubs look alike — check the label on each before sliding it on. VERIFIED

STAGE 6 — CONTINUED: SLEW SPOOL, RISER AND SLEEVE POSTS



1. Slew spool (P23)
2. Spool riser (P25) + 3 × #6 × 3/4in
3. 5/16in spool axle, 274 mm
4. Washer + jam nut in the Ø22 counterbore under the floor
5. Washer + nut on top (fitted with the wheel)
6. Sleeve posts (P26 ×2) + 2 × #6 × 3/4in each

Fig. 19 Spool axle, riser, spool and posts.



1. Lever axle + 3 hubs
2. Bearing blocks
3. Spool riser + slew spool
4. Sleeve posts
5. Box conduit fitting (bolted in Stage 7)

Fig. 20 Inside the box at the end of Stage 6 (top, rear and left side hidden).

YOU NEED (THIS PAGE)

P25	Spool riser	1
P23	Slew spool	1
H15	5/16in rod, 274 mm	1
H17	5/16in nut + SAE washer	1
P26	Slew sleeve post	2
H19	#6 × 3/4in wood screw	7
P18	Box conduit fitting (loose for now)	1

POST POSITIONS come from the model: post A at u −329, v −120; post B at u −274, v −216 (design frame). Each post's counterbore lines up with the spool's drum tangent. VERIFIED

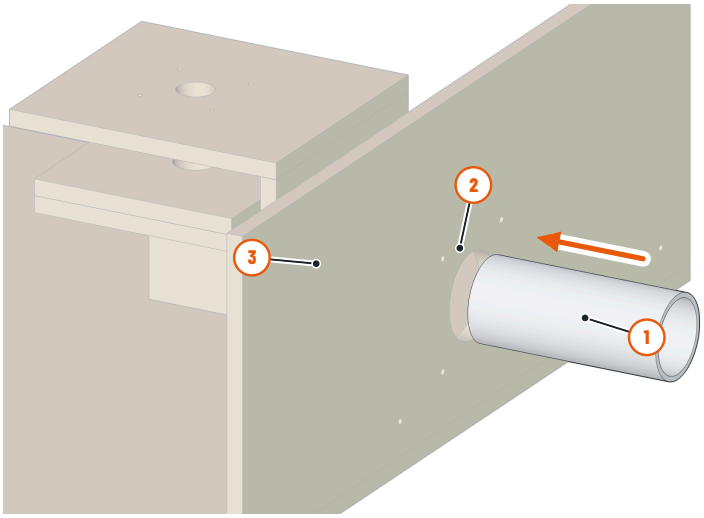
5. Push the 274 mm rod up through the floor's Ø8.5 hole from below. Fit a washer and a nut on its bottom end inside the Ø22 × 7 counterbore under the floor (the model uses a thin jam nut there so it stays flush).
6. Slide the riser down the rod onto the floor and screw it through its flange (3 × #6 × 3/4in).
7. Slide the spool onto the rod, drum end down, onto the riser. There is no washer between them: the gap is 0.5 mm. VERIFIED
8. Screw the two sleeve posts to the floor beside the spool, each with 2 × #6 × 3/4in through its base. Post **A** is the one nearer the levers. Each post's counterbore faces away from the spool.
9. Put the box conduit fitting (P18) aside — it goes in at Stage 7, together with the PTFE tubes.

CHECK. Each hub turns freely on the axle and the spool turns freely on its rod. Nothing in the box rubs the hubs.

STAGE 7

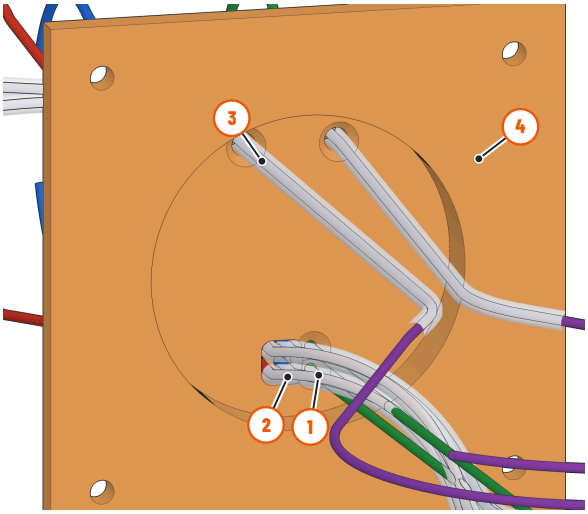
CONDUIT, PTFE TUBES AND BOX MOUNTING

The conduit links box and pedestal; the four arm tubes are seated in the box fitting, threaded through to the turret, then the box is bolted on.



1. 2in sch 40 PVC, 141 mm
2. Sandbox wall –u (Ø62 hole)
3. Pedestal fitting socket (12 mm deep)

Fig. 21 Conduit into the pedestal fitting through the sandbox wall.



1. 4 PTFE tubes seat together in the rounded window
2. Boom rope holes (bare rope)
3. Slew sleeve holes (Ø4.6, pass-through)
4. Conduit socket

Fig. 22 Box fitting P18 from the conduit side, with the tube and rope paths.

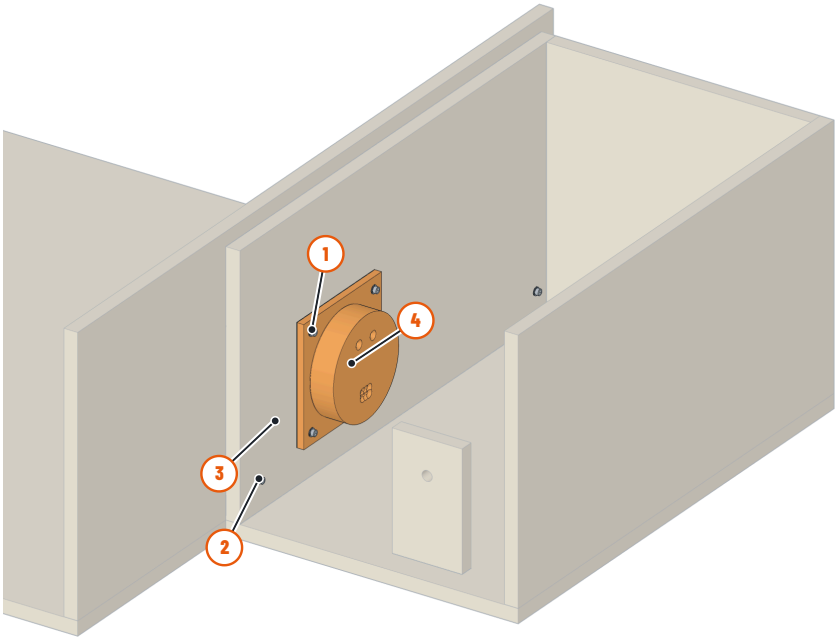
YOU NEED

S08	2in PVC conduit, 141 mm	1
P18	Box conduit fitting	1
S04	PTFE: stick_hi, stick_lo, bucket_hi, bucket_lo	4

WHY THIS ORDER. The tubes stop in P18 on the conduit side, and P18 is clamped between the box front and the sandbox wall by the same bolts. Once those bolts are in, the window cannot be reached. DERIVED

1. Push the conduit through the sandbox wall's Ø62 hole from outside, across the gap, through the pedestal wall's Ø62 hole and into P17's socket until it bottoms (12 mm deep). About 24 mm stays outside the sandbox wall. VERIFIED
2. Seat the box end of the four arm tubes in P18's rounded window — from the **conduit side** (the side with the big socket). They sit side by side and stop on the window floor.
3. Feed the four free ends out through the box front's Ø62 hole from inside the box, into the conduit's outer end, along the conduit, out through P17's window, into the copper elbow and up the PEX tube until they come out at the top of the turret.
4. Slide the box against the sandbox wall: the conduit end enters the box front's hole and P18's socket. P18 sits inside the box front with its socket on the conduit.

STAGE 7 — CONTINUED: BOLT THE BOX ON, SEAL THE CONDUIT



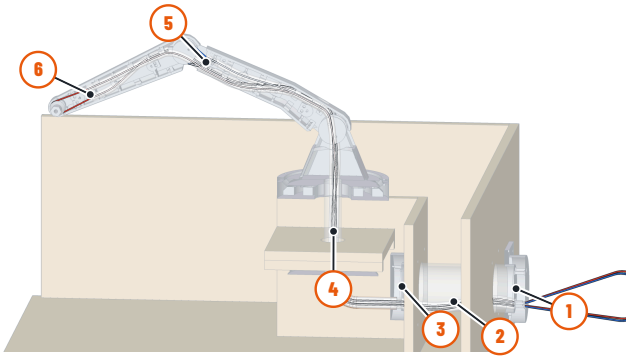
1. 4 × M3 × 35: fitting + box front + sandbox wall
2. 2 × M3 × 30 at the lower corners
3. Flat washers both sides; nuts inside the sandbox
4. Box conduit fitting (P18)

Fig. 23 Box against the sandbox wall: four fitting bolts and two corner bolts (box top, left side and sandbox walls hidden).

YOU NEED (THIS PAGE)

H06	M3 × 35 socket cap	4
H05	M3 × 30 socket cap	2
H08	M3 flat washer	12
H07	M3 nut	6
	Silicone	

5. Line up P18's four corner holes with the box front's and the sandbox wall's holes. Fit 4 × M3 × 35 from inside the box: flat washer under the head (on P18's flange), flat washer and nut on the sandbox side.
6. Fit the two lower-corner bolts: M3 × 30 through the box front and the sandbox wall, flat washers both sides.
7. Tighten all six evenly. Seal round the conduit at the outside of the sandbox wall with silicone.
8. At the turret, pull the four tubes up gently and leave them hanging out of the top for Stage 8. Leave their spare length coiled loosely in the conduit.



1. Box fitting window (tube stop)
2. Conduit: spare tube length coils here
3. Pedestal fitting window (tubes slide)
4. Up the copper elbow and the PEX tube
5. Stick tubes stop in the boom tip
6. Bucket tubes stop in the stick nose

Fig. 24 Where the four arm tubes run when finished.

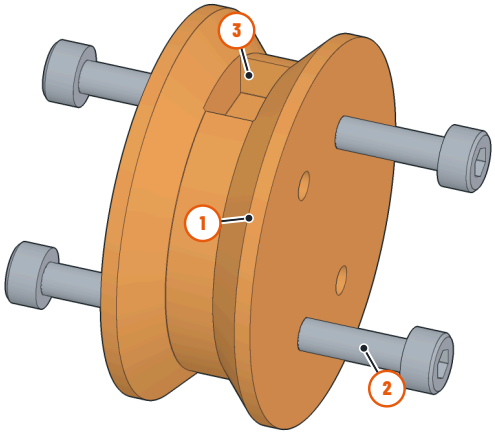
✓ **CHECK.** Each tube slides a few mm when pushed at the turret end, and does not come out of P18's window when pulled. The box stands flat on the same ground as the sandbox.

STAGE 8

ASSEMBLE THE ARM

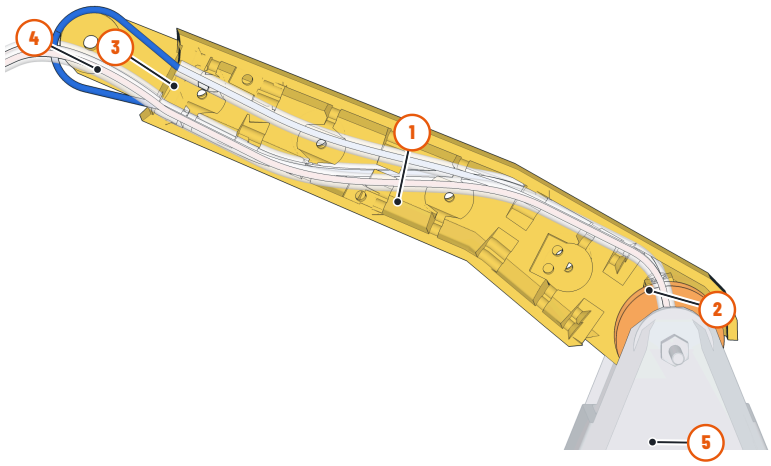
Boom, stick and bucket, each with its drum, its rope's joint end and its PTFE tubes laid in before the halves close.

READ APPENDIX D ITEMS 1-3 FIRST. The CAD shows three fit problems in this stage: the stick-drum key screws collide, two stick lug holes merge, and the joining-lug screws are enclosed once a member is closed. The steps below say where each one matters.



1. Boom drum (P05)
2. M3 × 12 key screw (4: two per face)
3. Crimp pocket (rope midpoint)

Fig. 25 Boom drum with its four key screws.



1. Boom half, right (P02), split face up
2. Boom drum: key-screw heads drop into the wall pockets
3. Stick tubes: to their stop in the boom tip
4. Bucket tubes: on through the boom tip
5. Boom ropes: straight down into the tower

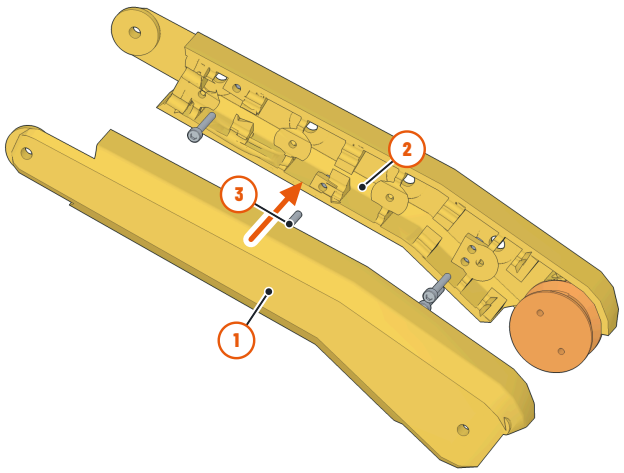
Fig. 26 Boom right half with drum, PTFE tubes and boom rope.

YOU NEED — BOOM

P05	Boom drum	1
H02	M3 × 12 socket cap (key screws)	4
S01	Boom rope 1384 mm (midpoint marked)	1
S02	Single crimp sleeve	1
P02	Boom half, right	1
P01	Boom half, left	1
H03	M3 × 20 socket cap (lugs)	4
H07	M3 nut	4

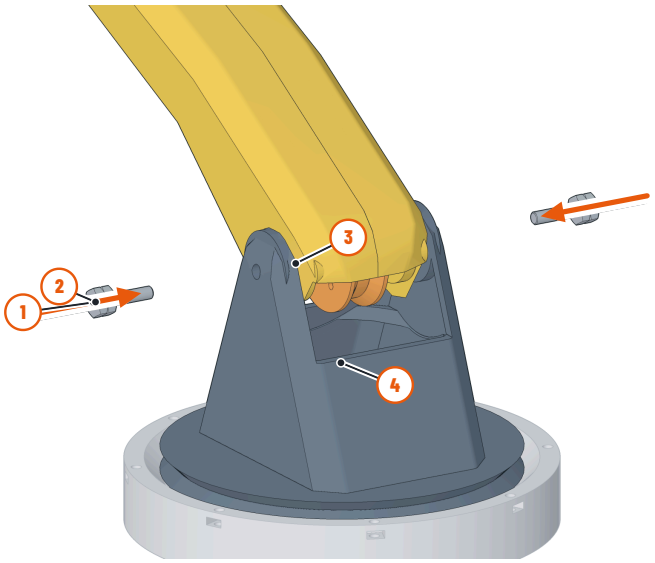
1. Screw two M3 × 12 into each face of the boom drum (4 screws, into the Ø2.6 pilots) until the underside of each head is **10 mm** from the drum face. The heads will sit in pockets in the boom walls. VERIFIED
2. Crimp the single sleeve onto the boom rope at its midpoint mark. Seat the sleeve in the drum's crimp pocket and wrap one tail each way round the groove.
3. Lay the drum in the right boom half (P02) so its two right-hand screw heads drop into the wall pockets at the boom root.
4. Lay the four PTFE tubes from the turret into P02's channels: the two **stick** tubes to their stop at the boom tip; the two **bucket** tubes on through the boom tip. Trim each stick tube so it seats on its stop.

STAGE 8 — CONTINUED: CLOSE THE BOOM AND PIN IT TO THE TOWER



1. Boom half, left (P01)
2. Boom half, right (P02)
3. Joining lugs: M3 × 20 + nut (see Open item 3)

Fig. 27 Closing the boom.



1. 8-32 rod, 42 mm (one per side)
2. Jam pair: 2 × 8-32 nut, outside the tower cheek
3. Inner end flush with the boom's inner wall
4. Tower cheek

Fig. 28 Boom split pins: one short rod per side.

5. Feed both boom rope tails down into the tower and the PEX tube, and push them on down to the box (see the tip).
6. Close the boom: put the left half (P01) on, the drum's left screw heads in its pockets.
7. Joining lugs: 4 × M3 × 20 + nut clamp the halves. Open item 3: in the CAD the lug screws cannot be reached once the halves are closed — decide how you will hold the halves before this step.
8. Set the boom root between the tower cheeks. Push a 42 mm 8-32 rod through each cheek into the boom wall until its inner end is flush with the boom's inner wall.
9. Fit two 8-32 nuts on the outer end of each pin and lock them against each other (jam pair) against the cheek.

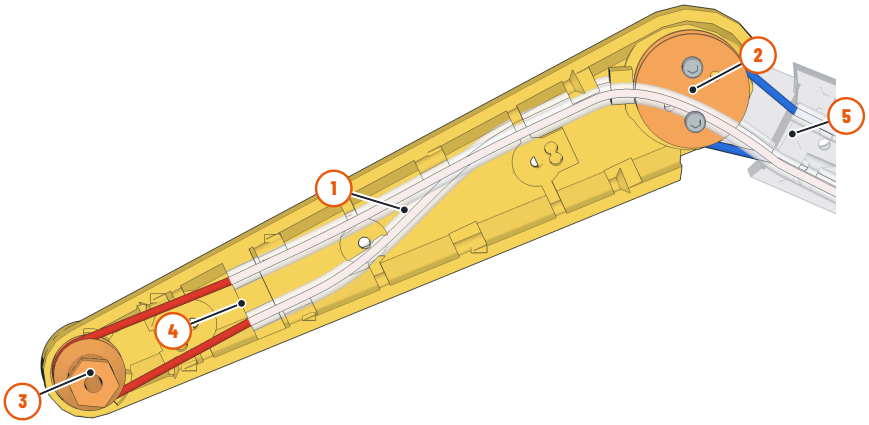
YOU NEED — BOOM PINS

H10	8-32 rod, 42 mm	2
H13	8-32 nut	4

★ **THREADING A BARE TAIL DOWN THE ELBOW.** Both fittings have bell-mouth entries on the conduit side. If a boom tail will not find its Ø2.4 hole at the bottom of the elbow, thread it up from the box side instead (through P18, the conduit, P17, the elbow and the PEX), take it over the drum and send the other tail down.

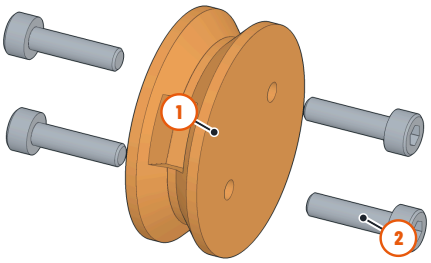
✕ **PIN TOO LONG INSIDE.** A split pin pushed past the boom's inner wall blocks the PTFE tubes that cross the joint axis. The inner ends must be flush. VERIFIED

STAGE 8 — CONTINUED: STICK



1. Stick half, right (P04)
2. Stick drum (P06) on its key screws
3. Bucket drum-axle (P07): journals in the nose holes
4. Bucket tubes: stop in the stick nose
5. Stick tube stops (boom tip)

Fig. 29 Stick right half: drum, bucket drum-axle, tubes and ropes.

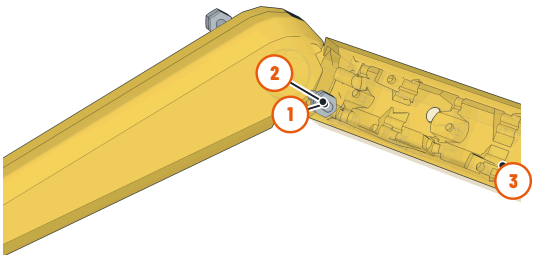


1. Stick drum (P06)
2. Key screws enter the same through-pilot from both sides

Fig. 30 Stick-drum key screws.

YOU NEED — STICK

P06	Stick drum	1
H02	Key screws — see Open item 1	4
S01	Stick rope 1821 mm	1
S02	Single crimp sleeve	2
S01	Bucket rope 2173 mm	1
P07	Bucket drum-axle	1
P04	Stick half, right	1
P03	Stick half, left	1
H03	M3 × 20 + nut (lugs)	3–4
H11	8-32 rod, 22 mm	2
H13	8-32 nut	4

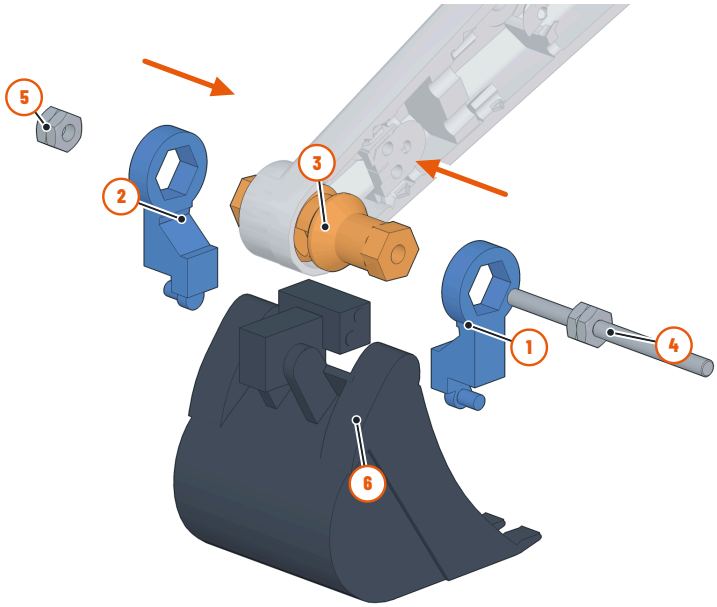


1. 8-32 rod, 22 mm (one per side)
2. Jam pair outside the boom tip
3. Boom tip

Fig. 31 Stick split pins.

1. Stick drum key screws: each pilot goes right through the 10 mm drum, and the model puts an M3 × 12 into it from each side with the heads 6.2 mm from the faces — the tips then overlap by 1.6 mm. Use M3 × 8 (in your kit) or shorten the M3 × 12 by 2 mm. OPEN ITEM
2. Crimp the stick rope's single sleeve at its midpoint, seat it in the stick drum's pocket and wrap the tails. Lay the drum in P04 with its screw heads in the pockets.
3. Push each stick tail into the end of its stick tube at the boom tip and on down the tube to the box.
4. Bucket drum-axle: put its journals in the nose holes of P04. Crimp the bucket rope's sleeve at its midpoint, seat it in the drum-axle's pocket, wrap the tails, and push them into the bucket tubes (laid in P04's channels to their stop in the nose) and on to the box.
5. Close with P03. Lugs: the two at the stick root overlap (2.9 mm apart) — only one screw fits there (Open item 2).
6. Fit the stick to the boom tip with a 22 mm 8-32 rod each side (inner ends flush) and a jam pair outside.

STAGE 8 — CONTINUED: BUCKET



YOU NEED — BUCKET

P08	Bucket	1
P09	Bucket ear, left (2026-09-29 file)	1
P10	Bucket ear, right (2026-09-29 file)	1
H12	8-32 rod, 65 mm (bucket pin)	1
H13	8-32 nut	4
	CA or epoxy	

GLUE LAST. Glue the ears only with the bucket pin fitted and tightened, so the hex sockets, the pin and the bucket lugs line up.

EARS SWAPPED. The left ear (P09) goes on the +v end of the drum-axle. Swapped ears do not seat on the bucket lugs.

1. Ear, left (P09): hex socket on the axle's hex end
2. Ear, right (P10)
3. Bucket drum-axle (P07) in the stick nose
4. 8-32 rod, 65 mm: the bucket pin
5. Jam pair each end
6. Bucket (P08): glue faces on the lugs

Fig. 32 Bucket, ears and bucket pin, exploded along the pin.

1. Dry fit: slide each ear's hex socket onto the drum-axle's hex end (left ear on the +v end). The ears' lower blocks rest on the bucket's lugs and the small pin under each ear goes into the bucket.
2. Push the 65 mm rod through the left ear, the drum-axle's bore and the right ear. Centre it.
3. Fit a jam pair on each end, snug against the ears: the rod clamps the ears to the drum-axle's shoulders.
4. When the bucket hangs straight and swings freely, glue each ear's lower block to its bucket lug (CA or epoxy), as on the original EX-MA. Let it cure fully before rigging.
5. Swing the bucket, stick and boom by hand through their ranges: nothing may rub.

CHECK — JOINT RANGES. Boom -40° … +45°, stick -45° … +50°, bucket -55° … +85° from the modelled pose, with no rubbing. The model is contact-free well beyond these (boom −42…+80°, stick −54…+150°, bucket −56…+170°). VERIFIED

STAGE 9

RIG THE ARM ROPES AT THE LEVERS

Set every joint to the middle of its range and every lever upright, then loop each tail onto its anchor and tension it.

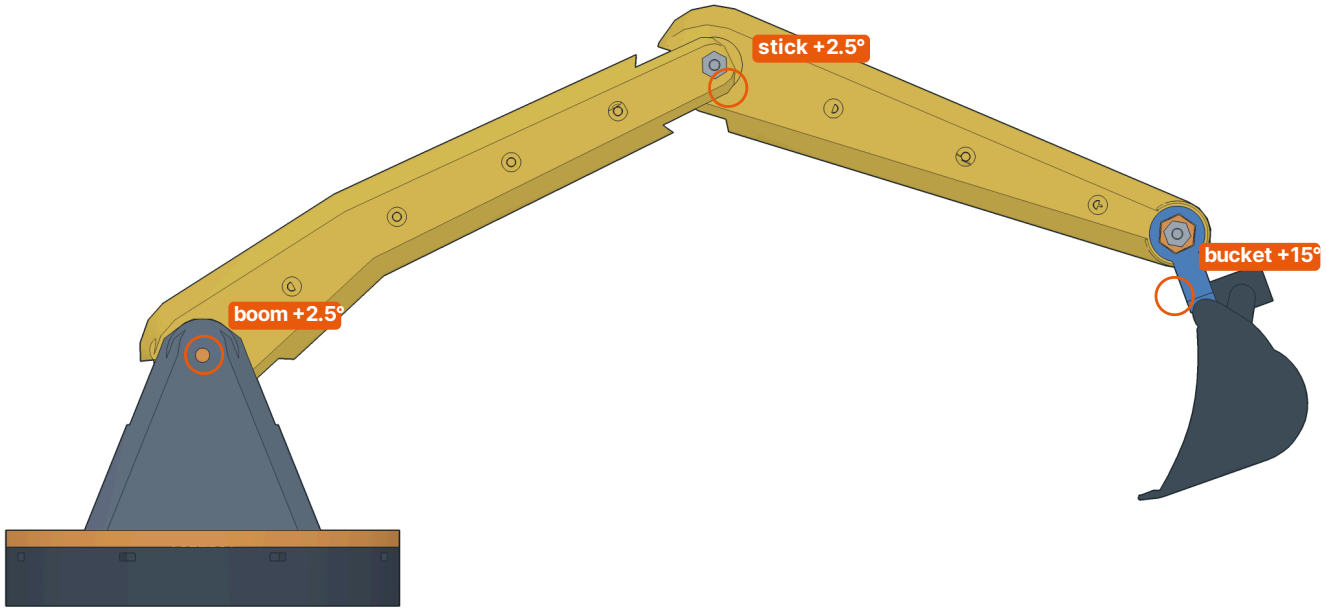


Fig. 33 The rigging pose: each joint at the middle of its working range, measured from the modelled pose (+ = raised).

CONFIGURATION: ZERO POSITIONS VERIFIED

Joint	Working range	Rig at
Boom	−40° … +45°	+2.5°
Stick	−45° … +50°	+2.5°
Bucket	−55° … +85°	+15°
All levers	slot end to slot end	upright
Slew	−90° … +90°	arm straight out, wheel centred

Crimping at mid-range leaves the same rope wrap on the joint drum at both ends of travel. Prop the arm in this pose with a block while you work. VERIFIED

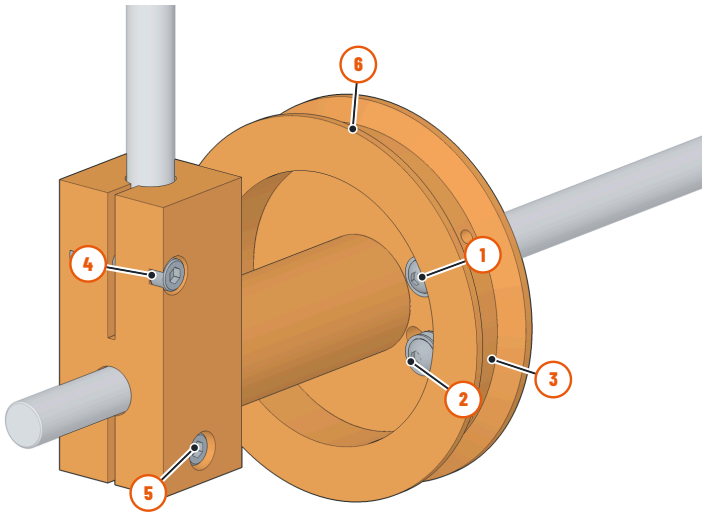
YOU NEED

S03	Double crimp sleeve	6
H01	M3 × 10 button head (in the hubs)	6
H08	M3 flat washer	12
H09	M3 split-lock washer	6
H07	M3 nut	6

1. Inside the box each tail comes out of P18's box face. Take each strand to its own lever drum: boom tails to the boom hub, stick to the stick hub, bucket to the bucket hub. Strands never cross.
2. Wrap the tail into the groove and pass it through the rim hole beside its anchor slot (upper strand to the upper anchor, lower to the lower).
3. With the anchor screw at the **outer** end of its slot (nearest the rope hole), pull the rope just taut and crimp a loop with a double sleeve so that it sits on the screw.
4. Stack on the screw: flat washer — rope loop — flat washer; the split-lock washer and nut on the other side of the web. Snug only.

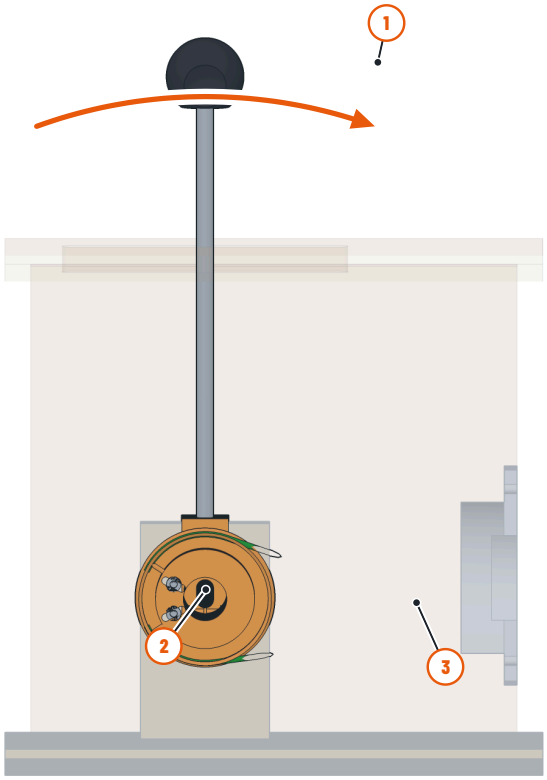
CRIMPING WITH THE JOINT OFF-CENTRE. A crimp made with the joint at one end of its range leaves no rope wrap at the other end, and the lever stops short. Re-check the rigging pose before every crimp.

STAGE 9 — CONTINUED: TENSION AND DIRECTION



1. Rope anchors: M3 × 10 button head in the slot (tension adjuster)
2. Washer — rope loop — washer under the head
3. Rope holes through the rim into the groove
4. Handle socket + pinch clamp (M3 × 20)
5. Drag clamp (M3 × 20 + lock washer)
6. Ø60 drum groove

Fig. 34 Lever hub (boom): anchors, rope holes and clamps.



1. Push = forward (toward the sandbox)
2. Lever axle (5/16in rod)
3. Boom rope strands to the box fitting

Fig. 35 Boom lever from the side (box cut away): push = forward.

5. Tension: slide each anchor screw along its slot **toward the drum centre** (away from the rope hole) until both strands are taut with the joint still at mid-range and the lever upright, then tighten the nut. Each slot gives about 8 mm of take-up.
6. Repeat for the other strand of the same circuit so the lever stays upright when both are taut.
7. Direction: push the lever forward (toward the sandbox). It must give boom down, stick out, bucket dump. If one is reversed, undo that rope at the joint drum and wrap it the other way round the drum.
8. Move each lever slowly end to end and watch its joint: smooth, no slack strand, the rope stays in every groove.

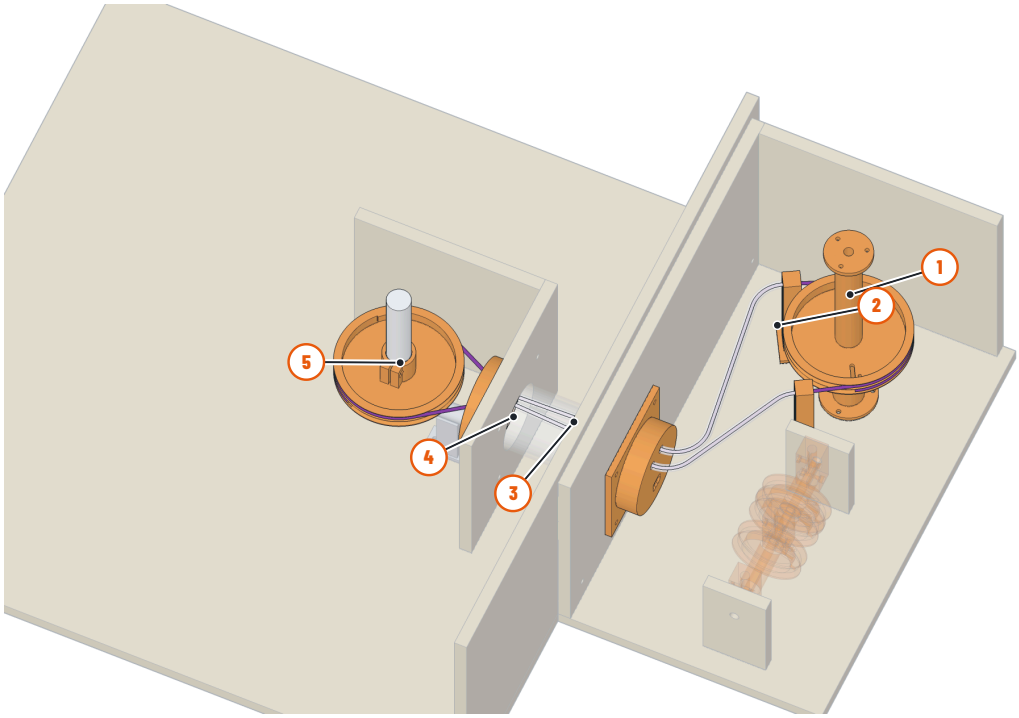
? **OPEN ITEM 7 — WHICH WAY TO SLIDE.** The checklist says “slide the anchors outward”. In the CAD the rope enters each slot at its **outer** end, so moving the screw inward is what takes up rope. Trust what you see: the strand tightens as the screw moves away from the rope hole. VERIFIED

✓ **CHECK — NO COUPLING.** Hold the bucket lever still and move the boom and stick levers through their full travel: the bucket must not move. The model predicts at most 0.66 mm of rope movement from tube bending (“Bowden coupling”). VERIFIED

STAGE 10

RIG THE SLEW ROPES

Two ropes in two PTFE sleeves: each one ends in a pocket on the pedestal drum and in a loop on the spool.



1. Slew spool (wheel)
2. Sleeve posts: sleeve ends seat here
3. PTFE sleeves through the box fitting and conduit
4. Pedestal fitting: angled counterbores (sleeve stops)
5. Slew drum: crimp pockets

Fig. 36 Slew path from above (box, pedestal and sandbox walls cut away): spool → posts → sleeves → conduit → pedestal fitting → slew drum.

YOU NEED

S04	PTFE slew sleeve A 339 mm, B 407 mm	2
S01	Slew rope A 829 mm, B 900 mm	2
S02	Single crimp sleeve	2
S03	Double crimp sleeve	2
H01	M3 × 10 button head (in the spool)	2
H08	M3 flat washer	4
H09	M3 split-lock washer	2
H07	M3 nut	2

1. Push sleeve A from its post (the one nearer the levers) through P18's upper slew hole on that side, along the conduit and into P17's angled counterbore until it stops. Seat its other end in post A's counterbore and trim the 10 mm extra. Same for sleeve B.
2. Crimp a single sleeve on one end of rope A. Thread the other end through the pedestal-fitting exit, sleeve A and post A to the spool.
3. Seat rope A's crimp in the **upper**-lane pocket of the pedestal drum and wrap it half a turn in that lane. Rope B: lower lane.

✕ **SWAPPED LANES.** Rope A (the shorter, 829 mm) runs in the **upper** lane on both drums and in sleeve A (339 mm). If ropes cross between the drums, the arm turns the wrong way.

STAGE 10 — CONTINUED: DRUM POSITION, SPOOL LOOPS, TENSION

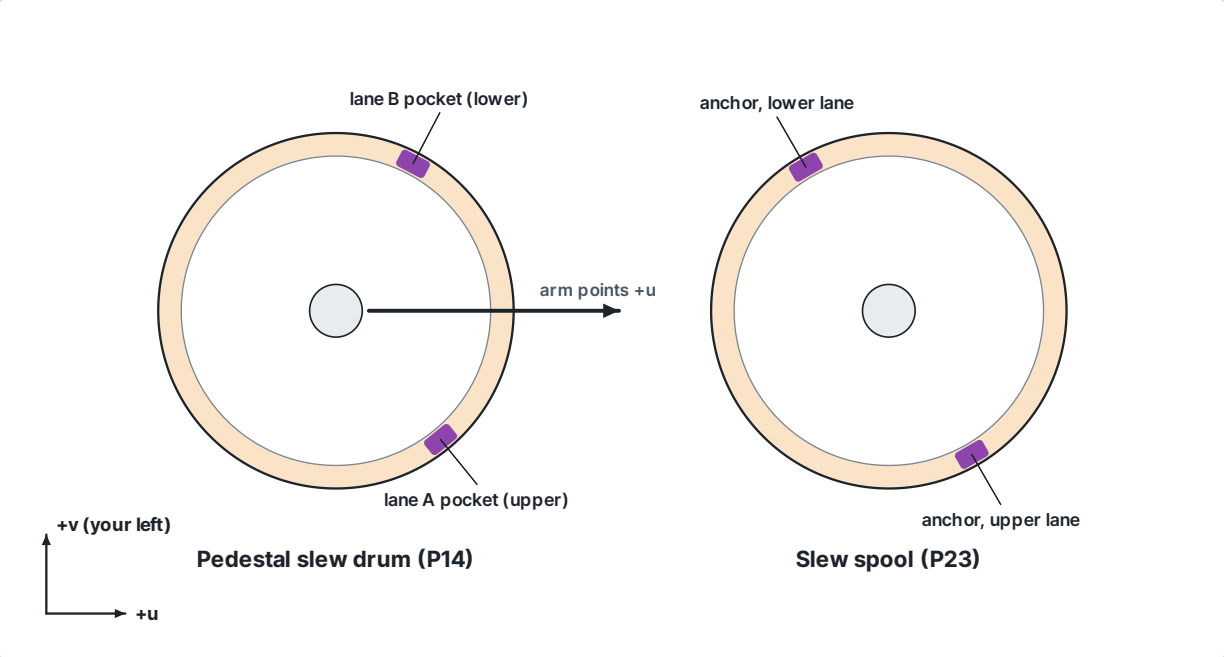
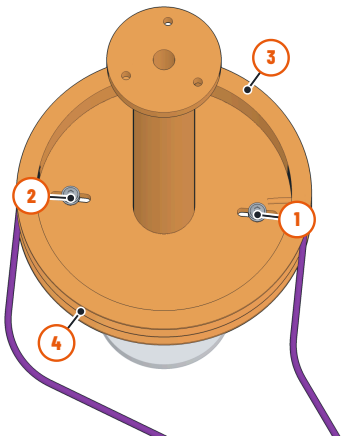


Fig. 37 Zero position seen from above, arm pointing straight out (+u): pocket and anchor angles from the model. VERIFIED

4. Point the arm straight out (+u, away from the box). Turn the pedestal drum on the PEX tube until its pockets are at the angles in the diagram (lane A 51° to the right of the arm, lane B 62° to the left), then tighten its pinch clamp hard. The drum must not slip on the tube. VERIFIED
5. Turn the spool so its two anchors are at the angles in the diagram (wheel centred).
6. Wrap each rope half a turn round its spool lane, pass it through the rim hole by its anchor, and crimp a loop (double sleeve) onto the anchor screw with the screw at the outer end of the slot: washer — loop — washer, lock washer and nut below the web.
7. Tension both ropes by sliding the anchors toward the spool centre, keeping the arm straight out. Each slot gives about 12 mm.
8. Turn the wheel left: the arm must swing to your left, 1 : 1, up to 90°. Then right to 90°.



1. Anchor A: M3 × 10 button head, washers each side of the loop
2. Anchor B
3. Upper lane
4. Lower lane

Fig. 38 Spool anchors and lanes from above.

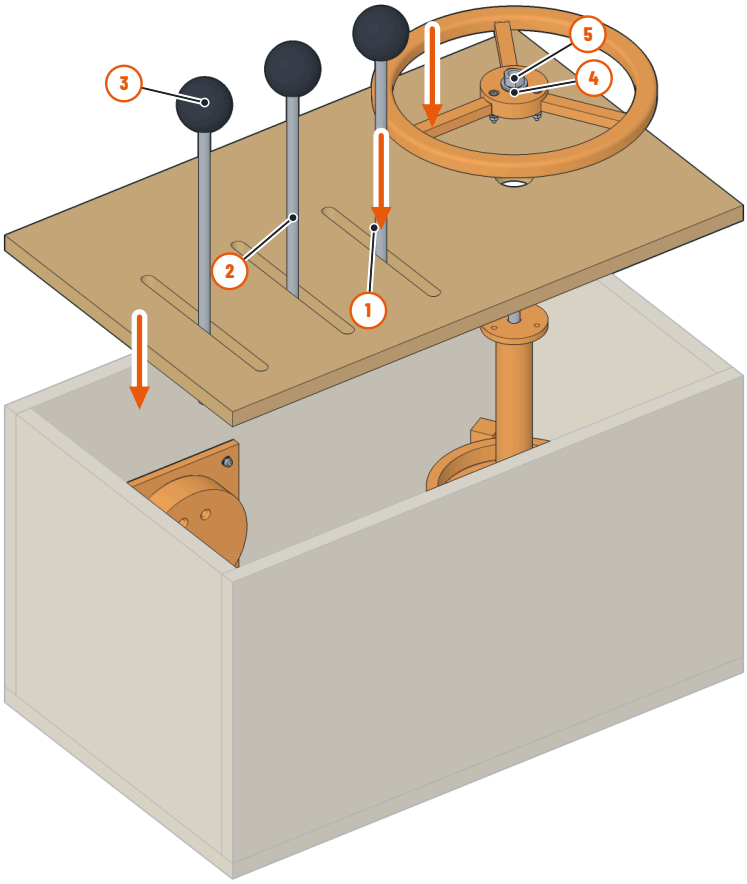
✓ **CHECK — WRAP LEFT AT THE ENDS.** At ±90° each rope must still have at least 90° of wrap on both drums (the model has 90–270°). If a rope runs out early, the drum or the spool was not centred: re-centre and re-crimp. VERIFIED

! **DRUM CLAMP.** A slipping pedestal drum feels like "the wheel turns but the arm doesn't". Tighten the M3 × 25 pinch clamp before tensioning, and re-check it after the first test.

STAGE 11

TOP PANEL, HANDLES, KNOBS AND WHEEL

The top panel's slots are the lever end stops. Fit it, push the handles through, and bolt the wheel on.



YOU NEED

W06	Box top (3 slots, 1 hole)	1
H20	#6 × 1-1/4in wood screw	16
H16	5/16in rod handle, 230 mm	3
P22	Lever knob	3
H03	M3 × 20 (handle clamps)	3
H03	M3 × 20 (drag clamps)	3
H09	M3 split-lock washer	9
H07	M3 nut	9
P24	Slew wheel	1
H03	M3 × 20 (wheel)	3
H17	5/16in nut + washer (spool top)	1
	CA or epoxy	

1. Top panel (W06)
2. 5/16in rod handle, 230 mm (×3)
3. Lever knob (P22 ×3), glued
4. Slew wheel (P24): 3 × M3 × 20 + lock washer + nut
5. 5/16in washer + nut on the spool axle

Fig. 39 Top panel, handles, knobs and wheel.

1. Lower the top onto the box: the spool tube passes through its Ø24.4 hole (the top is the spool's upper bearing). Screw it to the front, rear and both sides, 4 screws per edge — screws only, no glue, so it lifts off for service.
2. Push each handle rod down through its slot into its hub socket (25 mm deep). Tighten the handle pinch clamp (M3 × 20, lock washer and nut).
3. Glue a knob on each handle: the rod goes 14 mm into the knob's bore.
4. Bolt the wheel to the spool's top flange: 3 × M3 × 20, heads in the wheel's counterbores, lock washer and nut under the flange.
5. Fit the washer and nut on top of the spool axle. Snug it until the spool has no up-down play but still turns freely. ASSUMPTION
6. Fit the three drag-clamp screws (M3 × 20, lock washer, nut) loosely; they are set in Stage 12.

✓ **CHECK.** Each lever stops on the end of its slot at both ends of travel — the slot is the hard stop. The model has the handle touch the slot end exactly at the end of travel, with nothing else touching. VERIFIED

STAGE 12

FINAL TEST AND CONFIGURATION

Run every control through its full range, check for coupling, then set the lever friction.

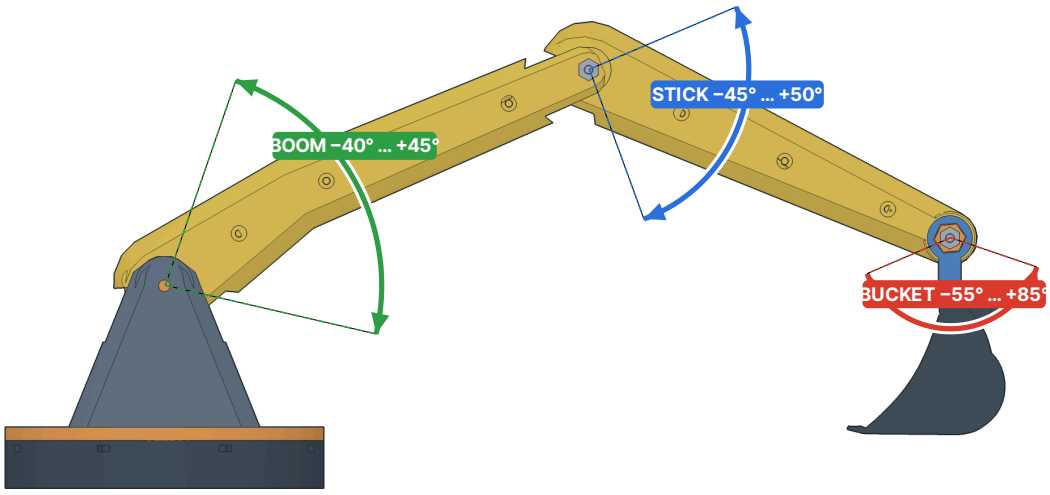


Fig. 40 Working ranges set by the lever slots, measured from the modelled pose shown (+ = raised).

FUNCTION TEST VERIFIED

1. Each lever, slowly, end to end: its joint moves through its whole working range and the handle stops on the slot ends.
2. Wheel to −90° and +90°: the arm follows 1 : 1 and no rope jumps its groove.
3. No coupling: move the boom and the stick through their ranges while watching the bucket. It must not move.
4. At every end stop, look at the ropes: each stays in its groove, and no PTFE tube is pinched at a joint.
5. Turn the turret by the wheel through ±90°: smooth, no knocking from the ball ring.

CONFIGURATION: LEVER HOLDING FRICTION

6. Drag clamps: with the arm extended, tighten each lever's drag-clamp screw a little at a time until the lever holds its position when you let go. The boom needs the most; the bucket almost none.
7. Re-tension every circuit after the first hour of use (rope stretch and crimp seating), using the slotted anchors.

Lever	Holding torque (model)	Clamp force (μ 0.3)
Boom	0.65 N·m	172 N
Stick	0.23 N·m	60 N
Bucket	0.04 N·m	10 N

From the box validation report VERIFIED. Use them to compare levers: the boom clamp needs several times the stick's.

✓ **DONE.** When every check on this page passes, the build is complete. Keep the hardware bags: the spares are listed in Appendix A.

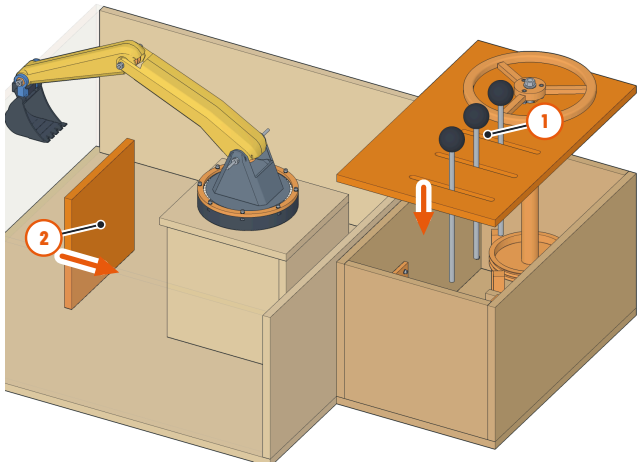
AFTER ASSEMBLY

OPERATION AND MAINTENANCE

OPERATING

- Stand behind the box. Left hand: boom and stick levers. Right hand: bucket lever and slew wheel.
- Move one control at a time at first. Push forward = boom down, stick out, bucket dump; wheel left = arm left.
- The slot ends are the stops. Do not lever against them or force the wheel past ±90°.
- Do not lift or carry the machine by the arm, the levers or the wheel. Carry the sandbox and the box together from below.
- Dig in dry sand only. Brush sand off the slewing ring and out of the lever slots after use.

SERVICE ACCESS VERIFIED



1. Box top: lift off (screws only)
2. Pedestal wall +u: unscrew (move the sand away first)

Fig. 41 The two service hatches: box top and pedestal wall +u.

- **Box top** — lift off (screws only): lever hubs, anchors, spool, sleeve posts, box fitting.
- **Pedestal wall +u** — unscrew (screws only): slew drum clamp, copper elbow, sleeve ends. Clear the sand from that side first.
- **An arm half** — opening one half gives access to its drum, crimp pocket and tubes. See Open item 3 for how the halves are held.

MAINTENANCE SCHEDULE ASSUMPTION

When	Do
After the first hour	Re-tension all four circuits (rope stretch and crimp seating). VERIFIED
Every session	Brush off sand; check the knobs and the handle clamps are tight.
Monthly	Check the jam nuts on the five joint pins and the 8 retaining-ring screws; check the slew drum clamp.
If a lever feels slack	Re-tension at its anchors (Stage 9).
If the arm drifts	Tighten that lever's drag clamp (Stage 12).

REPLACING A ROPE DERIVED

1. Lift the box top; undo the two loops of that circuit at its lever drum.
2. Open the arm half that holds its joint drum and lift the crimp out of the pocket.
3. Pull the old rope out from the arm end; use it to measure the new one (cut lengths in Appendix C).
4. Re-rig as in Stages 8 and 9 (or 10 for slew).

AFTER ASSEMBLY

TROUBLESHOOTING

Find the symptom, check the likely cause, then go to the stage named in the fix. Causes come from how the circuits are built; where a cause is general practice it is not tagged separately.

Symptom	Likely cause	Fix
A lever moves the wrong joint	Strands crossed in the box, or ropes swapped at the box fitting	Each circuit's two strands go to its own hub; the box fitting columns run boom / stick / bucket in the same order as the hubs (Stage 9).
A joint moves the wrong way	Rope wrapped the wrong way round the joint drum	Reverse that rope's wrap on its joint drum (Stage 9, step 7).
Moving one lever moves another joint	PTFE tube not seated at the box fitting or at its arm stop; tube pinched	Re-seat both tube ends; check the tube is not trapped by a split pin or lug (Stage 8).
Lever stops short of the slot end	Rope crimped with the joint off-centre; rope too tight	Return to the rigging pose and re-crimp (Stage 9).
Lever floppy, joint lags	Rope slack	Slide the anchors toward the drum centre (Stage 9, step 5).
Arm sags when you let go	Drag clamp too loose	Tighten that lever's drag clamp (Stage 12, step 6).
Lever very stiff	Drag clamp too tight; axle nuts squeezing the hubs; hub rubbing a neighbour	Ease the drag clamp; back the axle nuts off; hubs need their 0.5 mm gaps.
Rope jumps out of a groove	Slack rope at an end stop	Re-tension; check the joint is not driven past its working range.
Wheel turns, arm does not (or slips)	Pedestal slew drum or turret-hub clamp slipping on the PEX tube	Tighten both M3 × 25 pinch clamps (Stages 5 and 10).
Arm turns the opposite way to the wheel	Slew ropes crossed between lanes	Rope A upper lane on both drums, rope B lower (Stage 10).
Slew range less than ±90°	Drum or spool not centred when the ropes were crimped	Re-centre both (Stage 10, steps 4–5) and re-crimp.
Turret rocks or lifts	Retaining ring too high	Shim under the retaining ring (Stage 5 check).
Turret binds or knocks	Missing or extra BB; uneven ring screws; sand in the race	57 balls; loosen the ring screws evenly; clean the race.
Bucket wobbles on its pin	Jam nuts loose; ear glue joint failed	Re-lock the jam pairs; re-glue the ear with the pin fitted.
Handle turns in its hub	Handle pinch clamp loose	Tighten the M3 × 20 handle clamp (Stage 11).
Sand coming out at the box	Conduit seal missing at the sandbox wall	Seal round the conduit with silicone outside the wall (Stage 7).
Rope frays at a fitting	Burr on the copper or the PVC end	Deburr; check the rope runs through the bell-mouth holes.

APPENDIX A

BILL OF MATERIALS

From `2026-09-24-bill-of-materials.md` VERIFIED; base and ear files updated to the 2026-09-29 rebuilds.

PRINTED PARTS

ID	Part	Qty	Material	Plate
P01	Boom half, left (+v)	1	PETG	arm 1
P02	Boom half, right (−v)	1	PETG	arm 1
P03	Stick half, left (+v)	1	PETG	arm 1
P04	Stick half, right (−v)	1	PETG	arm 1
P05	Boom drum (Ø30 pitch)	1	PLA+	other 6
P06	Stick drum (Ø26 pitch)	1	PLA+	other 6
P07	Bucket drum-axle (Ø17 pitch)	1	PLA+	other 6
P08	Bucket	1	PLA+	other 6
P09	Bucket ear, left (G)	1	PLA+	other 6 (new file)
P10	Bucket ear, right (H)	1	PLA+	other 6 (new file)
P11	Tower (turret)	1	PLA+	other 4
P12	Base (lower race)	1	PLA+	other 4 (new file)
P13	Slewing-ring retaining ring	1	PLA+	other 5
P14	Pedestal slew drum (Ø110)	1	PLA+	other 1
P15	PEX bushing	1	PLA+	other 1
P16	Elbow support	1	PLA+	other 1
P17	Conduit fitting, pedestal end	1	PLA+	other 1
P18	Conduit fitting, box end	1	PLA+	other 3
P19	Lever hub, boom	1	PLA+	other 2
P20	Lever hub, stick	1	PLA+	other 2
P21	Lever hub, bucket	1	PLA+	other 3
P22	Lever knob (Ø36)	3	PLA+	other 2
P23	Slew spool (Ø110)	1	PLA+	other 3
P24	Slew wheel (Ø184)	1	PLA+	other 2
P25	Spool riser	1	PLA+	other 2
P26	Slew sleeve post	2	PLA+	other 2

HARDWARE

ID	Item	Qty	Used for
H01	M3 × 10 button head	8	rope-tail anchors on the 3 lever hubs (2 each) and the slew spool (2)
H02	M3 × 12 socket head cap	16	boom + stick drum key screws (8), retaining ring (8)
H03	M3 × 20 socket head cap	17	joining lugs (8), drag clamps (3), handle clamps (3), slew wheel (3)
H04	M3 × 25 socket head cap	6	PEX pinch clamps (2), pedestal conduit fitting (4)
H05	M3 × 30 socket head cap	2	box front to sandbox wall, lower corners
H06	M3 × 35 socket head cap	4	box conduit fitting through box front + sandbox wall
H07	M3 hex nut	45	with the screws above
H08	M3 flat washer	36	fittings, box bolts, rope anchors
H09	M3 split-lock washer	27	under nuts/heads on moving or clamping joints
H10	8-32 rod, 42 mm	2	boom split pins
H11	8-32 rod, 22 mm	2	stick split pins
H12	8-32 rod, 65 mm	1	bucket pin
H13	8-32 hex nut	12	jam pairs on the pins
H14	5/16in rod, 205 mm	1	lever axle
H15	5/16in rod, 274 mm	1	slew spool axle
H16	5/16in rod, 230 mm	3	lever handles
H17	5/16in nut + SAE washer	4	2 on the lever axle, 2 on the spool axle
H18	#8 × 1in flat-head wood screw	4	base to the pedestal top
H19	#6 × 3/4in flat-head wood screw	16	bearing blocks, riser, posts, bushing, elbow support
H20	#6 × 1-1/4in flat-head wood screw	120	panel joints, 4 per edge, with wood glue

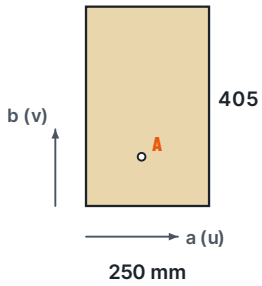
M3 counts all fit in the Fgruh kit. Spares left: 8-32 rod about 711 mm and 10 nuts; #4-40 and #6-32 screws unused.

APPENDIX B

DRILLING DRAWINGS

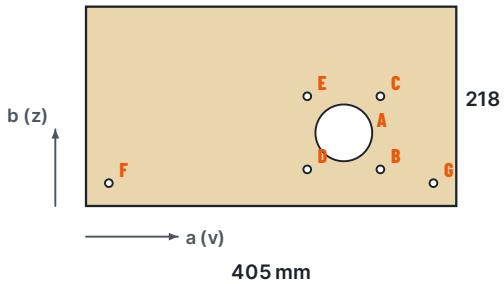
Every drilled panel, drawn to scale from `build-box.py` VERIFIED. Positions in mm from the panel's lower-left corner as drawn: **a** along the horizontal edge, **b** up the vertical edge. The box floor's Ø8.5 hole also gets a Ø22 × 7 counterbore from below.

W01 box floor — seen from above; a from the −u edge, b from the −v edge (operator's right)



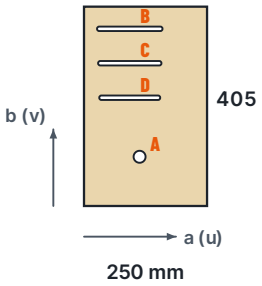
	Size	a	b	Purpose
A	Ø8.5	113.0	100.0	slew spool axle (nut + washer in a Ø22 x 7 counterbore from below)

W02 box front — seen from the +u side; a from the −v edge (operator's right), b from the bottom edge



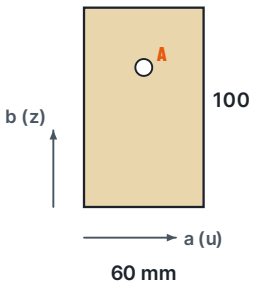
	Size	a	b	Purpose
A	Ø62.0	282.0	80.0	2in conduit
B	Ø4.6	322.0	40.0	fitting bolt (M3 × 35 through the sandbox wall)
C	Ø4.6	322.0	120.0	fitting bolt (M3 × 35 through the sandbox wall)
D	Ø4.6	242.0	40.0	fitting bolt (M3 × 35 through the sandbox wall)
E	Ø4.6	242.0	120.0	fitting bolt (M3 × 35 through the sandbox wall)
F	Ø4.6	25.0	25.0	box-to-sandbox bolt (M3 × 30)
G	Ø4.6	380.0	25.0	box-to-sandbox bolt (M3 × 30)

W06 box top (removable) — seen from above; a from the −u edge, b from the −v edge (operator's right)



	Size	a	b	Purpose
A	Ø24.4	113.0	100.0	spool tube bearing
B	slot 133.0 × 9.0	93.0	360.0	boom lever slot = hard stop (42 deg swing)
C	slot 128.7 × 9.0	93.0	290.0	stick lever slot = hard stop (41 deg swing)
D	slot 123.9 × 9.0	93.0	220.0	bucket lever slot = hard stop (40 deg swing)

W07 + W08 axle bearing blocks (×2, identical) — seen from the −v side; a from the −u edge, b from the bottom edge

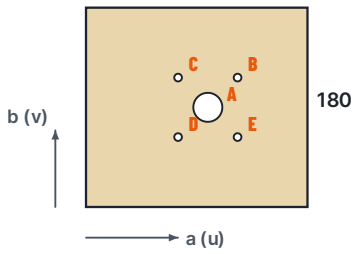


	Size	a	b	Purpose
A	Ø8.5	30.0	70.0	5/16in lever axle

APPENDIX B

DRILLING DRAWINGS (CONTINUED)

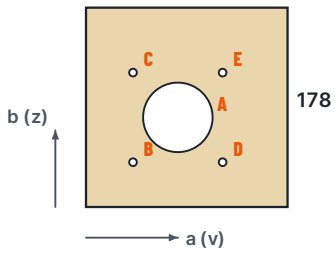
W09 pedestal top — seen from above; a from the –u edge, b from the –v edge (operator's right)



200 mm

	Size	a	b	Purpose
A	Ø26.2	110.0	90.0	PEX turret tube clearance
B	Ø3.0	136.9	116.9	pilot, #8 screw from the base
C	Ø3.0	83.1	116.9	pilot, #8 screw from the base
D	Ø3.0	83.1	63.1	pilot, #8 screw from the base
E	Ø3.0	136.9	63.1	pilot, #8 screw from the base

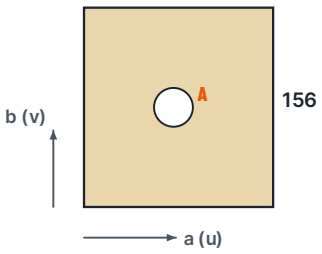
W10 pedestal wall -u (conduit) — seen from the +u side; a from the –v edge (operator's right), b from the bottom edge



180 mm

	Size	a	b	Purpose
A	Ø62.0	82.0	80.0	2in conduit
B	Ø4.6	42.0	40.0	fitting bolt (M3 × 25)
C	Ø4.6	42.0	120.0	fitting bolt (M3 × 25)
D	Ø4.6	122.0	40.0	fitting bolt (M3 × 25)
E	Ø4.6	122.0	120.0	fitting bolt (M3 × 25)

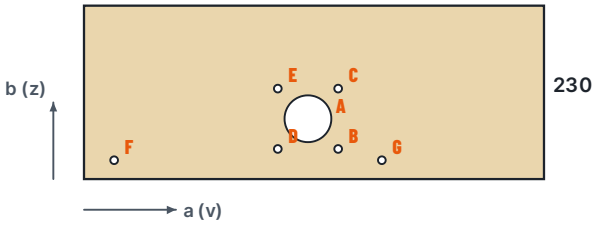
W14 pedestal shelf — seen from above; a from the –u edge, b from the –v edge (operator's right)



148 mm

	Size	a	b	Purpose
A	Ø30.4	70.0	78.0	slew-tube-bushing

W18 sandbox wall -u (box side) — seen from the +u side; a from the –v edge (operator's right), b from the bottom edge



610 mm

	Size	a	b	Purpose
A	Ø62.0	297.0	80.0	2in conduit (seal with silicone)
B	Ø4.6	337.0	40.0	bolt from the control box
C	Ø4.6	337.0	120.0	bolt from the control box
D	Ø4.6	257.0	40.0	bolt from the control box
E	Ø4.6	257.0	120.0	bolt from the control box
F	Ø4.6	40.0	25.0	bolt from the control box
G	Ø4.6	395.0	25.0	bolt from the control box

APPENDIX C

ROPE, TUBE AND PLATE REFERENCE

CUT LENGTHS VERIFIED

Piece	Cut (mm)	Modelled path (mm)
Boom rope	1384	1264 + 2 × 60 loops
Stick rope	1821	1701 + 2 × 60
Bucket rope	2173	2053 + 2 × 60
Slew rope A	829	769 + 60
Slew rope B	900	840 + 60
PTFE stick_hi	647	637 + 10 trim
PTFE stick_lo	658	648 + 10
PTFE bucket_hi	813	803 + 10
PTFE bucket_lo	822	812 + 10
Slew sleeve A	339	329 + 10
Slew sleeve B	407	397 + 10

Modelled lengths from `Validation/2026-09-24-box-report.md`; cut lengths from the bill of materials.

ROPE TRAVEL PER CIRCUIT VERIFIED

Circuit	Range	Rope travel per side	Control
Boom	85°	22.3 mm	lever 42°
Stick	95°	21.6 mm	lever 41°
Bucket	140°	20.8 mm	lever 40°
Slew	180°	172.8 mm	wheel 180°

LEVER ROPES IN THE BOX VERIFIED

Strand	Length	Fleet	Wrap over swing
boom upper	118	11.3°	62–105°
boom lower	118	11.2°	52–95°
stick upper	116	4.5°	63–104°
stick lower	116	4.4°	53–94°
bucket upper	117	9.8°	63–103°
bucket lower	118	9.8°	54–94°

SLEW WRAP VERIFIED

Slew	Rope A spool / pedestal	Rope B spool / pedestal
−90°	90° / 270°	270° / 90°
0°	180° / 180°	180° / 180°
+90°	270° / 90°	90° / 270°

SLICER PROJECTS

`3MF/2026-09-27-bambu-arm-shell-petg.3mf` · `3MF/2026-09-27-orca-arm-shell-petg.3mf`
`3MF/2026-09-27-bambu-other-parts-plaplus.3mf` · `3MF/2026-09-27-orca-other-parts-plaplus.3mf`

APPENDIX D

OPEN ITEMS AND VERIFICATION

Found while preparing this manual by checking the project's own CAD and hardware model. None of them is fixed in the project files; each needs your decision. OPEN ITEM

#	What the files show	Options
1	**Stick-drum key screws collide** The stick drum is 10 mm thick and each key pilot runs right through it. The hardware model puts one M3 × 12 into each pilot from each side with its head in a 3.5 mm wall pocket, which places the two tips at v −0.8 and +0.8 mm: they overlap by 1.6 mm (10.5 mm³ in the interference check). The boom drum (14 mm) does not have this problem: its screws stop 5 mm short of the centre.	Use M3 × 8 socket caps on the stick drum (in your kit; 1.8 mm in the drum), or grind 2 mm off each M3 × 12. Both keep the heads in their pockets.
2	**Stick root lugs overlap** Joining lugs 1 and 2 of the stick are 2.9 mm apart, so their Ø3.4 holes merge into one figure-of-eight hole and their screws and nuts overlap (interference check).	Fit one M3 × 20 in that lug; the stick then has 3 lug screws (1 spare).
3	**Joining-lug screws are enclosed** On both members every lug screw head sits inside the closed shell, and the outer walls have no access hole on the screw axis (ray test on the CAD). As modelled, the lug screws cannot be fitted or tightened once a boom or stick is closed.	Decide before Stage 8. Options: (a) drill a small access hole through the outer wall on each lug axis and fit the nut captive with a drop of CA — changes the exterior; (b) close the halves without lug screws and hold them with tape or a few dots of CA along the seam — loses easy opening for service; (c) add access in the CAD and reprint the halves.
4	**5/16in axle ends** The bill of materials puts 5/16in nuts on both ends of the lever axle and the spool axle. One design note calls the rod "smooth"; the validation report checks the axle both plain and threaded.	If your rod is threaded rod, nothing changes. If it is smooth, thread the ends with a 5/16-18 die or use 5/16in shaft collars.
5	**Pedestal not fixed to the floor** The cut list says "the pedestal and the elbow support screw to" the sandbox floor, but no screws for the pedestal are counted in the bill of materials or placed in the hardware model.	4 extra #6 × 1-1/4in up through the sandbox floor into the ±v walls (Stage 4).
6	**Slicer plates hold the old base and ears** Plates 4 and 6 of the PLA+ project were made before the 2026-09-29 rebuilds.	Replace them with STL/2026-09-29-base.stl and STL/2026-09-29-bucket-ear-{left,right}-cad.stl (Stage 1).
7	**Tension direction** The checklist says to slide the rope anchors "outward". In the CAD the rope enters each anchor slot at its outer end, so the strand tightens as the screw moves inward.	Slide toward the drum centre (Stages 9 and 10).

WHAT WAS VERIFIED, AND HOW

Illustrations	Rendered from the project STL files, the plywood panels and the hardware solids at their modelled positions. Base and ears use the 2026-09-29 files.
Hardware interference	Every pair of fasteners checked for overlap (manifold boolean). Found items 1 and 2; the only other overlap is the copper elbow inside the PEX tube, which the model draws as a solid rod (not a real clash).
Numbers	Dimensions, ranges and lengths are taken from `exma_common.py`, `build-box.py`, `build-parts.py`, the bill of materials, the checklist and the validation reports.
Not verified	Physical fit of real prints, glue strength, rope friction, how hard a person can push. Tips and items tagged ASSUMPTION are general practice.

APPENDIX E

OPTION: COMPACT CONTROL BOX

On 2026-09-28 a smaller box was specified (`2026-09-28-compact-control-box-spec.md`) without changing the arm, the slew or the reach. It is a specification only — **no CAD model was made**, so this manual's figures show the full-size box. VERIFIED

	Modelled box (this manual)	Compact box (spec)
Outside u × v × z	250 × 405 × 242	214 × 362 × 156
Lever handle length / rod	225 / 230 mm	140 / 145 mm
Handle spacing	70 mm	60 mm
Lever axle rod	205 mm	185 mm
Spool axle rod	274 mm	188 mm
Slew spool tube	145 mm	59 mm (reprint)
Lever hubs	as printed	boom + bucket reprinted (sleeve −10 mm)
Lid slots (boom / stick / bucket)	133 / 129 / 124	66.1 / 64.1 / 61.8
Slew sleeves A / B	339 / 407	330 / 397 (cut)
Knob stroke (boom)	177 mm	114 mm
Boom holding force at the knob	2.7 N	4.2 N

IF YOU CHOOSE THE COMPACT BOX cut its panels from the spec (floor and lid 214 × 362, front and rear 362 × 132, sides 190 × 132), reprint the slew spool and the boom and bucket hubs, and drill two new box-to-sandbox holes. The Stage 6 and Stage 11 steps are the same; only the dimensions change.

UNCHANGED Arm, turret, pedestal, conduit, fittings, drums, all arm ropes and the four arm PTFE tubes. The lever axle height and position stay the same, so the rope angles do not change. VERIFIED

APPENDIX F

FILES AND REVISION

PROJECT FILES USED

`STL/`	printed parts (P01–P26), print-ready
`STEP/`	solid models of the new parts, the wood, and the 2026-09-29 base, ring and ears
`3MF/`	slicer projects (Bambu Studio, OrcaSlicer)
`Source/`	build scripts; all parameters in `2026-09-24-exma_common.py`
`Validation/`	box, kinematics and wall-thickness reports
`2026-09-24-bill-of-materials.md`	quantities, plates, materials
`2026-09-24-cut-list.md`	plywood sizes and holes
`2026-09-25-print-and-assembly-checklist.md`	the checklist this manual expands
`2026-09-28-compact-control-box-spec.md`	Appendix E
`References/IMG_2449, IMG_2454`	photos of the M3 kit and the Hillman parts

THIS MANUAL'S SOURCE

`source/2026-10-10-manual-scene-export.py`	geometry → scene for the renderer
`source/2026-10-10-manual-figure-specs.py`	cameras, callouts, exploded offsets
`source/2026-10-10-manual-render.html`	three.js illustration renderer
`source/2026-10-10-manual-figures.mjs`	renders all figures
`source/2026-10-10-manual-svg.py`	overlays and vector diagrams
`source/2026-10-10-build-manual.py`	text and page layout
`source/2026-10-10-manual.css`	page styles
`source/2026-10-10-manual-print.mjs`	HTML → PDF
`source/2026-10-10-build-manual.sh`	runs the whole chain
`figures/`	rendered figures + callout data

REVISION

Date	Change
2026-10-10	First edition. Covers the modelled design with the 2026-09-29 base and ears.